

Accessing Toyota Event / Camera Data

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TOYOTA SAFETY SENSE

TOYOTA SAFETY SENSE (TSS / LSS)

- What is it? How does it work?

EVENT DATA ACCESSED THROUGH TOYOTA TECHSTREAM

- Vehicle Control History (VCH)
- VCH / FFD with IMAGES
- How to obtain the data using the Toyota Techstream software and Mongoose cable (or CDR 900)

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TOYOTA SAFETY SENSE (TSS / LSS)

TOYOTA SAFETY SENSE 3.0

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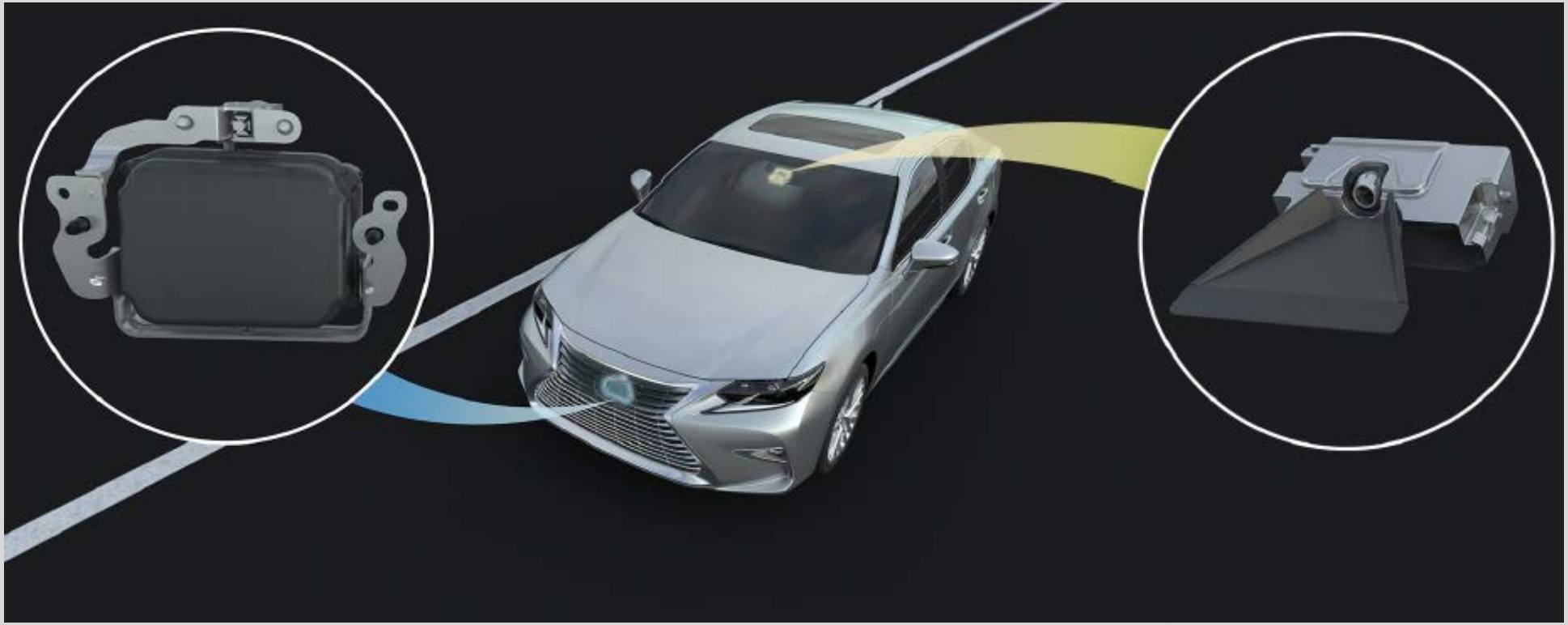
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TOYOTA SAFETY SENSE



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TOYOTA SAFETY SENSE



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TSS – Pre-Collision System (PCS)

What Does it Do?

- Front Millimeter Wave Radar and the front recognition camera are designed to:
  - Detect vehicles and pedestrians (TSS-P)
  - Cyclists (TSS 2.0 & TSS 2.5+)
  - Motorcycles (TSS 3.0)
- To help mitigate collisions, the system 1st activates an audible and visual alert in addition to Brake Assist if the brakes are applied by the driver
- If the driver does not brake in time this is followed by automated braking

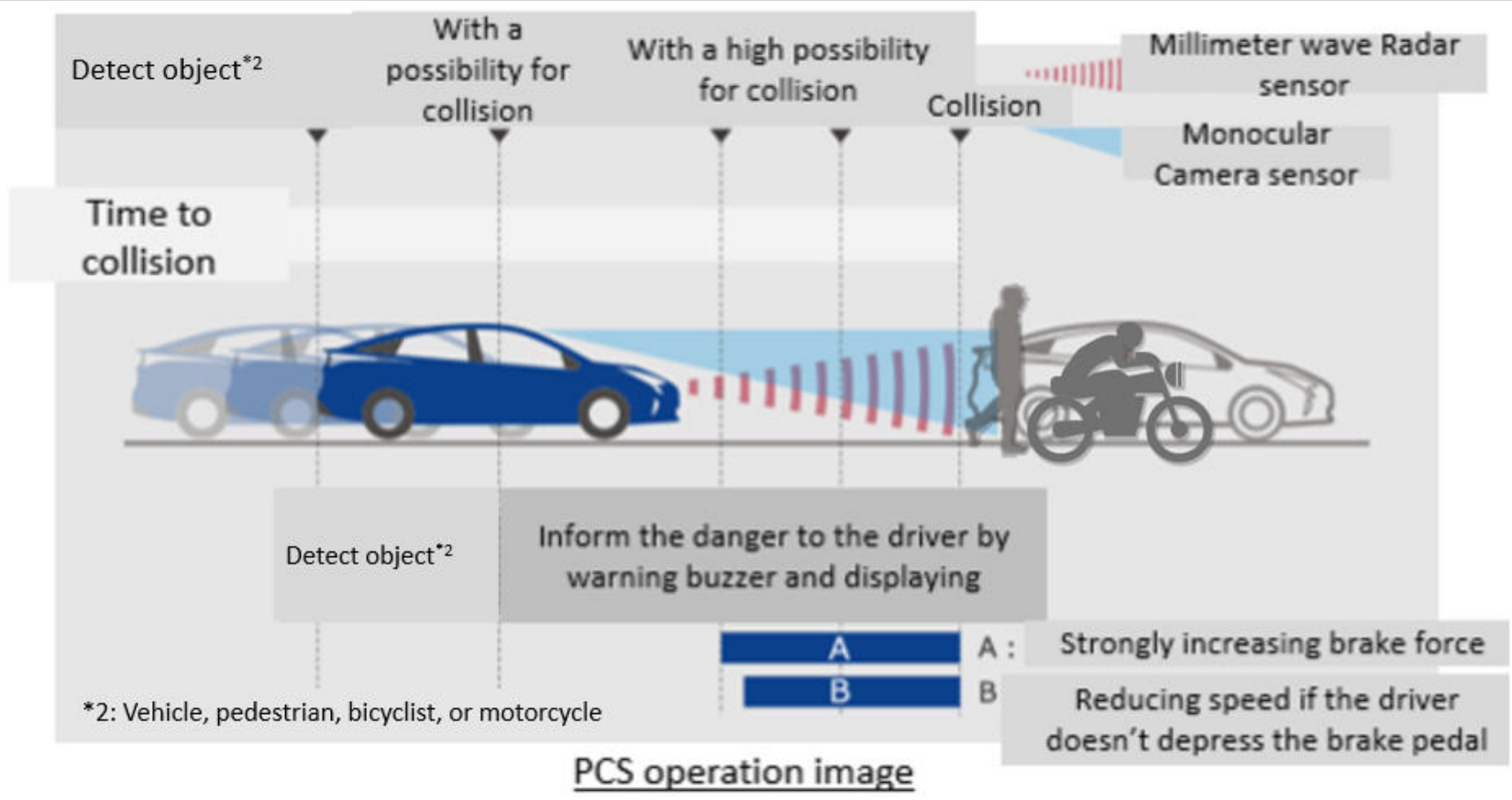
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TSS – Pre-Collision System (PCS)

How Does it Work?

- PCS utilizes a front recognition camera mounted above the rear-view mirror and a front millimeter wave radar sensor mounted in the grille of the vehicle
- Activation of the PCS system is based on Time To Collision (TTC), which is calculated from distance and relative velocity

TSS – Pre-Collision System (PCS)



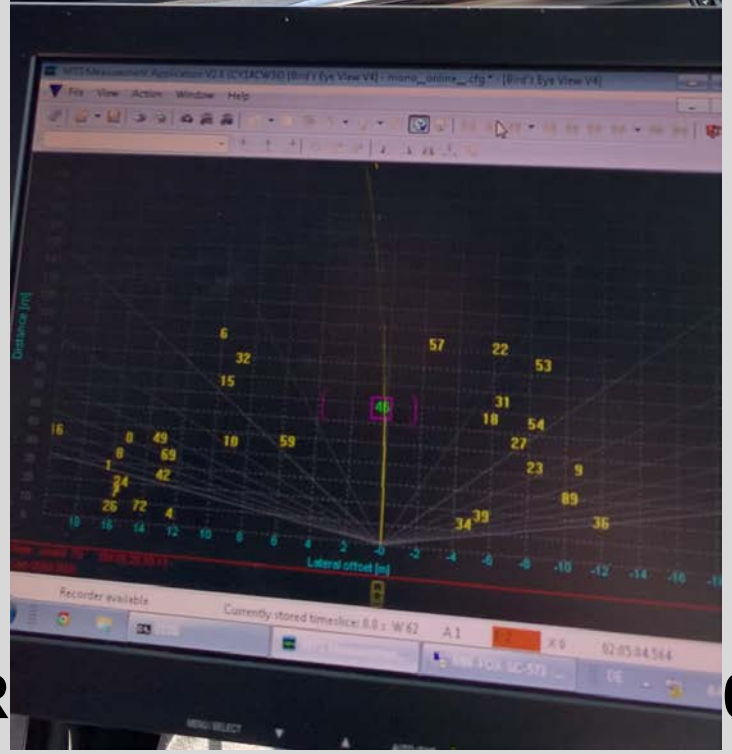
PCS operation image

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TSS – Pre-Colli CS)





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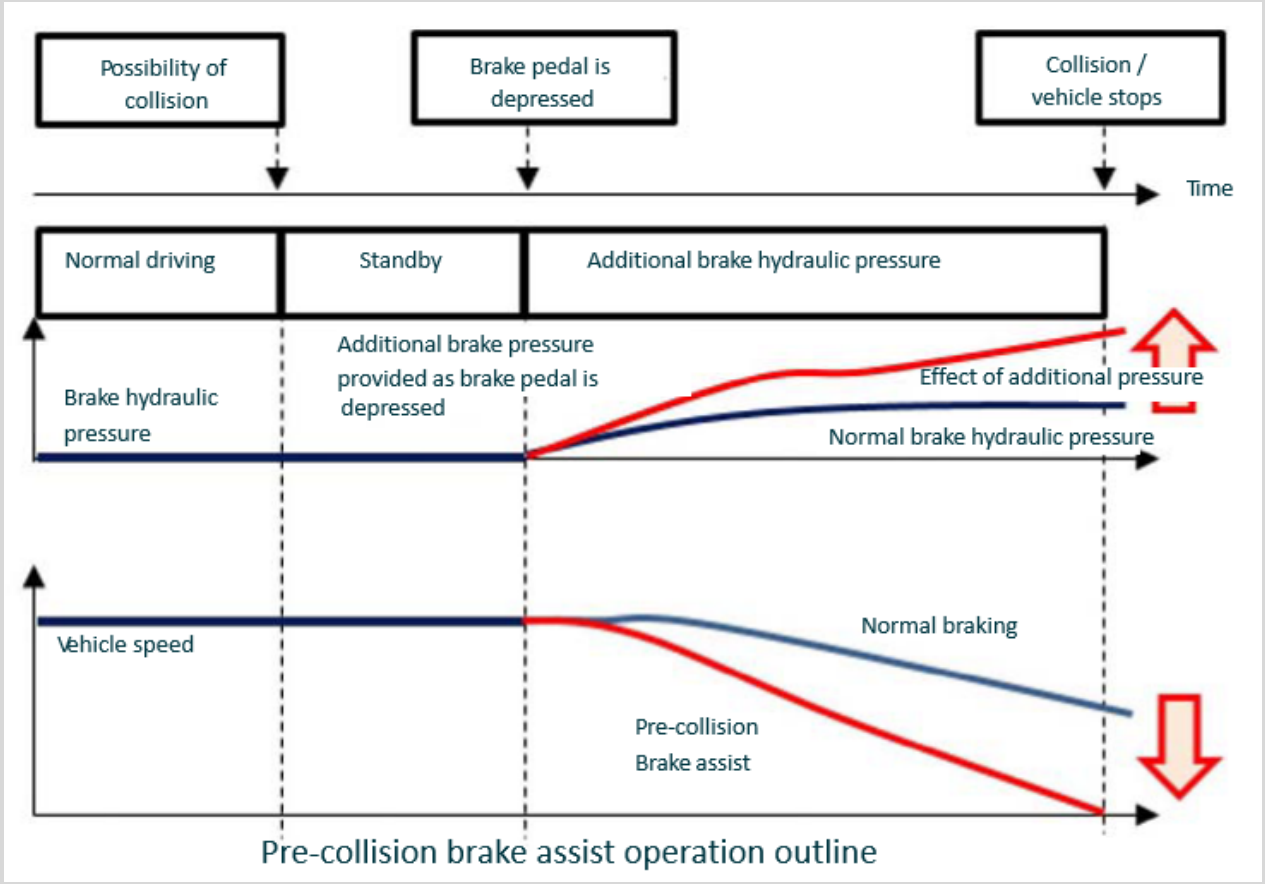
Pre-Collision Brake Alarm (ALM)



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Pre-Collision Brake Assist (PBA)

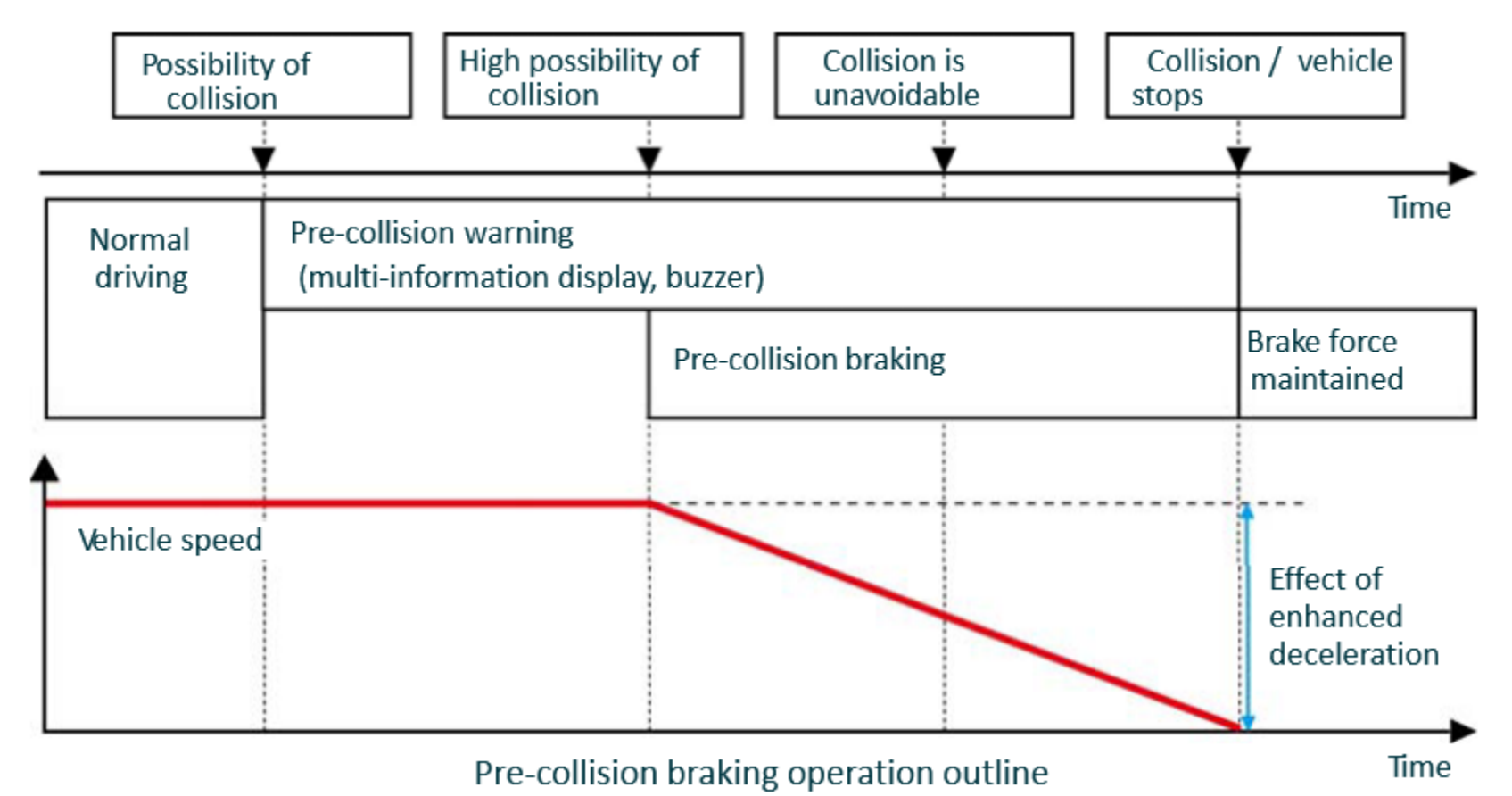


Pre-collision brake assist operation outline

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Pre-Collision Brake Assist (PB)



Pre-collision braking operation outline

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4 Generations of TSS

- 1st Generation – TSS-P & TSS-C
- 2nd Generation – TSS 2.0
- 3rd Generation – TSS 2.5 & 2.5+
- 4th Generation – TSS 3.0 & 3.0+

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1st Generation – TSS- P

Pre-Collision System

- The front millimeter wave **radar sensor** assembly and forward recognition camera detect the position and relative speed of preceding vehicles **and pedestrians**.

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1st Generation – TSS- P

2023 production vehicles include:

- 2023 Tacoma
- 2023 4Runner
- 2023 Lexus GX (LSS+)
- 2023 Lexus LC (LSS+ A)

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2nd Generation – TSS 2.0

- Pre-Collision System enhanced to better detect pedestrians in low light conditions
- Added Daytime Bicyclist Detection
- Also equipped with Lane Trace Assist

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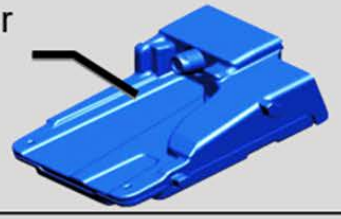
2nd Generation – TSS 2.0

2023 production vehicles include:

- 2023 Sienna

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2nd Generation – TSS 2.0

			TSS 2.0	TSS 1.0
Package			LEXUS Safety System+2.0	LEXUS Safety System+
Camera Design			Monocular Camera 	Monocular Camera 
Recognition Object	Pedestrian	at daytime	●	●
		at nighttime	●	—
	Bicycle		●	—
	White Line		●	●
	Roadside		●	—
	Lights Illumination		●	●
	Vehicle		●	●
	Road Sign		● [-N/A and -Other]	● [only -Other]

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2nd Generation – TSS 2.0

- Improved camera for PCS secures the same performance operation at daytime **and low-light conditions**



TSS



TSS 2.0

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3rd Gene

TOYOTA

TSS 2.0, TSS 2.5 and TSS 2.5+ Comparison Table

System			TSS 2.0	TSS 2.5	TSS 2.5+
Pre-Collision	Vehicle Detection		✓	✓	✓
	Pedestrian Detection		✓	✓	✓
	Cyclist Detection		✓	✓	✓
	Low-light Pedestrian Detection		✓	✓	✓
	Emergency Steering Assist (ESA)			✓ (Pedestrian Only)	✓ (Pedestrian, Cyclist, Vehicle)
	Front-to-front risk detection (oncoming vehicle)			X	✓
	Intersection Support	Left-Turn Oncoming Vehicle		✓	✓
		Left/Right-Turn Oncoming Pedestrian		✓	✓
Dynamic Radar Cruise Control	Turn Signal-linked Control		✓	✓	✓
	Curve Speed Management			X	✓
Lane Keeping	Lane Departure Alert with Steering Control		✓	✓	✓
	Lane Tracing Assist		✓	✓	✓
	Sway Warning System		✓	✓	✓
Lights	Automatic High Beam		✓	✓	✓

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3rd Generation – TSS 2.5 & 2.5+

2023 production vehicles include:

TSS 2.5

- RAV4 & RAV4 Hybrid
- Sequoia
- Tundra

TSS 2.5+

- Camry & Camry Hybrid
- Venza
- Highlander & Highlander Hybrid
- RAV4 Prime

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3rd Generation – TSS 2.5 & 2.5+

2023 production vehicles include:

LSS+ 2.5

- IS
- ES
- LS
- RC
- UX
- LX



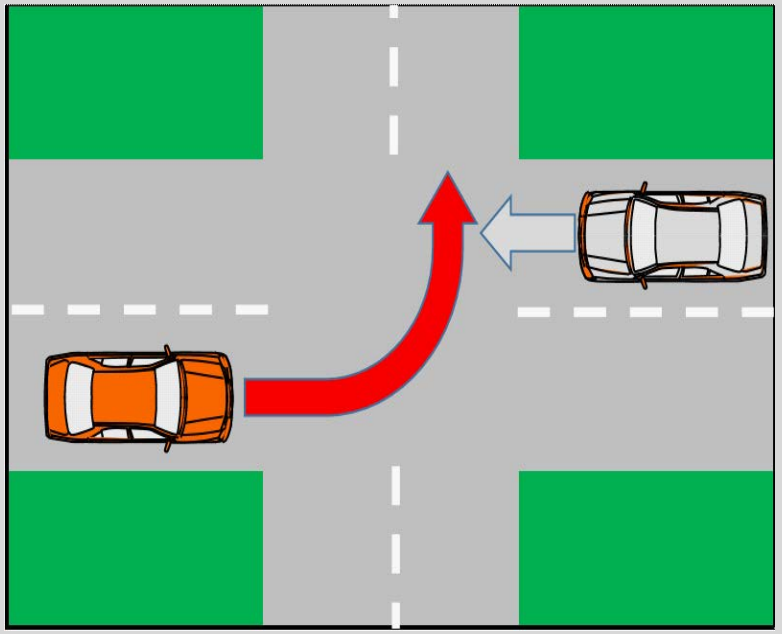
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3rd Generation – TSS 2.5 & 2.5+

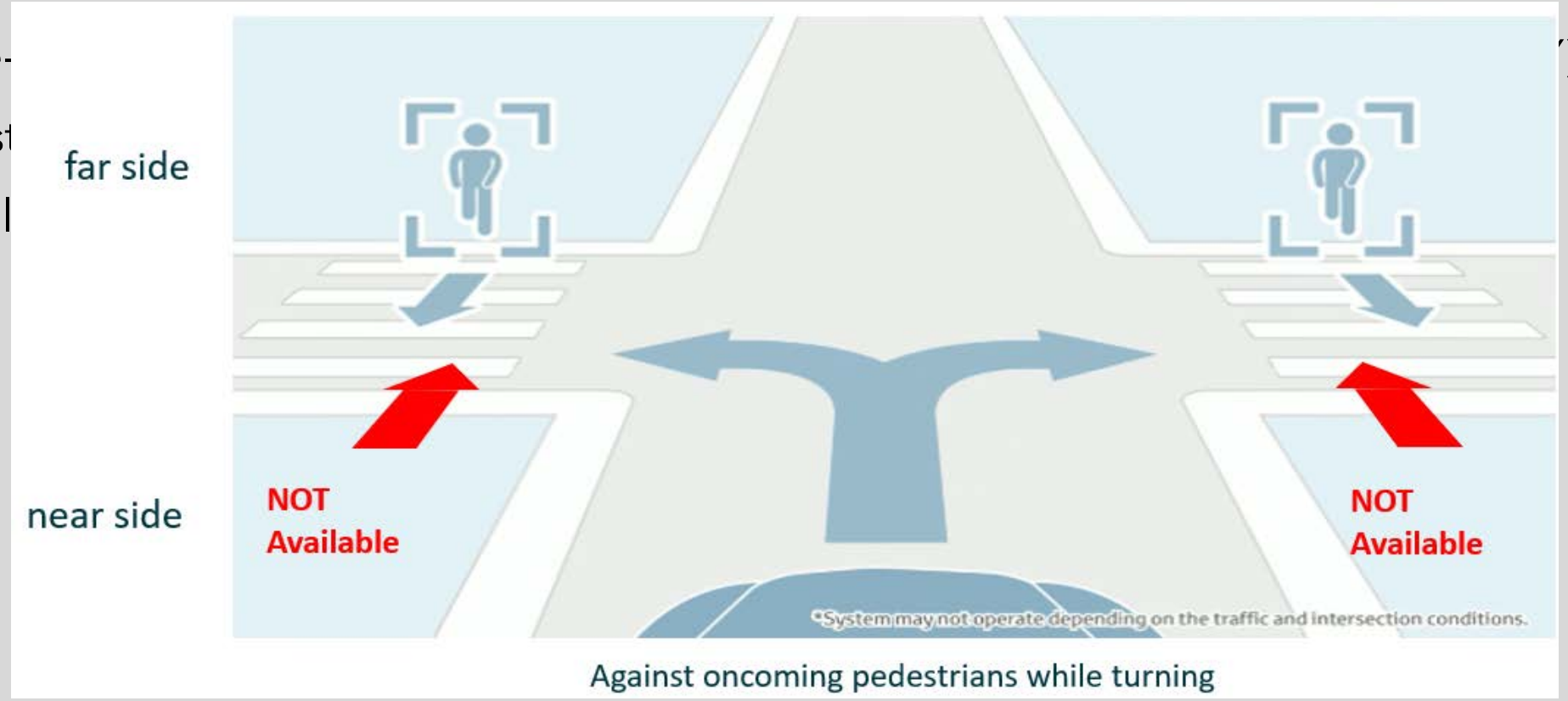
- Pre-Collision detection when turning across the path of a vehicle





3rd Generation – TSS 2.5 & 2.5+

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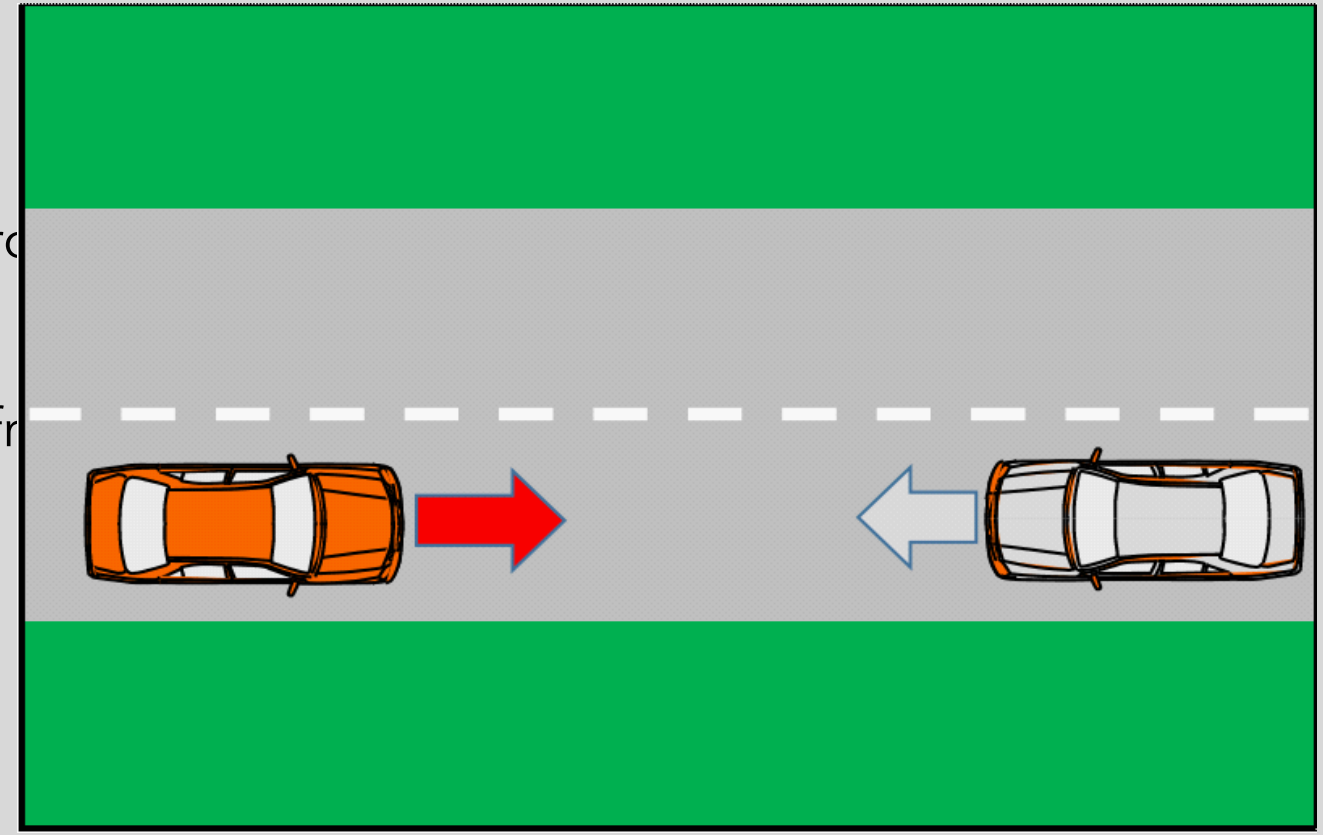
3rd Generation – TSS 2.5 & 2.5+

TSS 2.5+ (DENSO)

- PCS equipped with fr

TSS 2.5 (Continental)

- **NOT** equipped with fr



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4th Generation – TSS 3.0

- Enhanced camera and radar sensors
- Added Daytime Motorcycle Detection
- Added Crossing Vehicles for intersection collision avoidance support
- Acceleration Suppression Function (ASF)
- Added Proactive Driving Assist (PDA)
- Emergency Driving Stop System (EDSS) *
- Traffic Jam Assist *



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4th Generation – TSS 3.0

2023 production vehicles include:

- bZ4x
- Corolla & Corolla Hybrid & Corolla Hatchback
- Corolla Cross
- Crown
- NX
- RX
- RZ



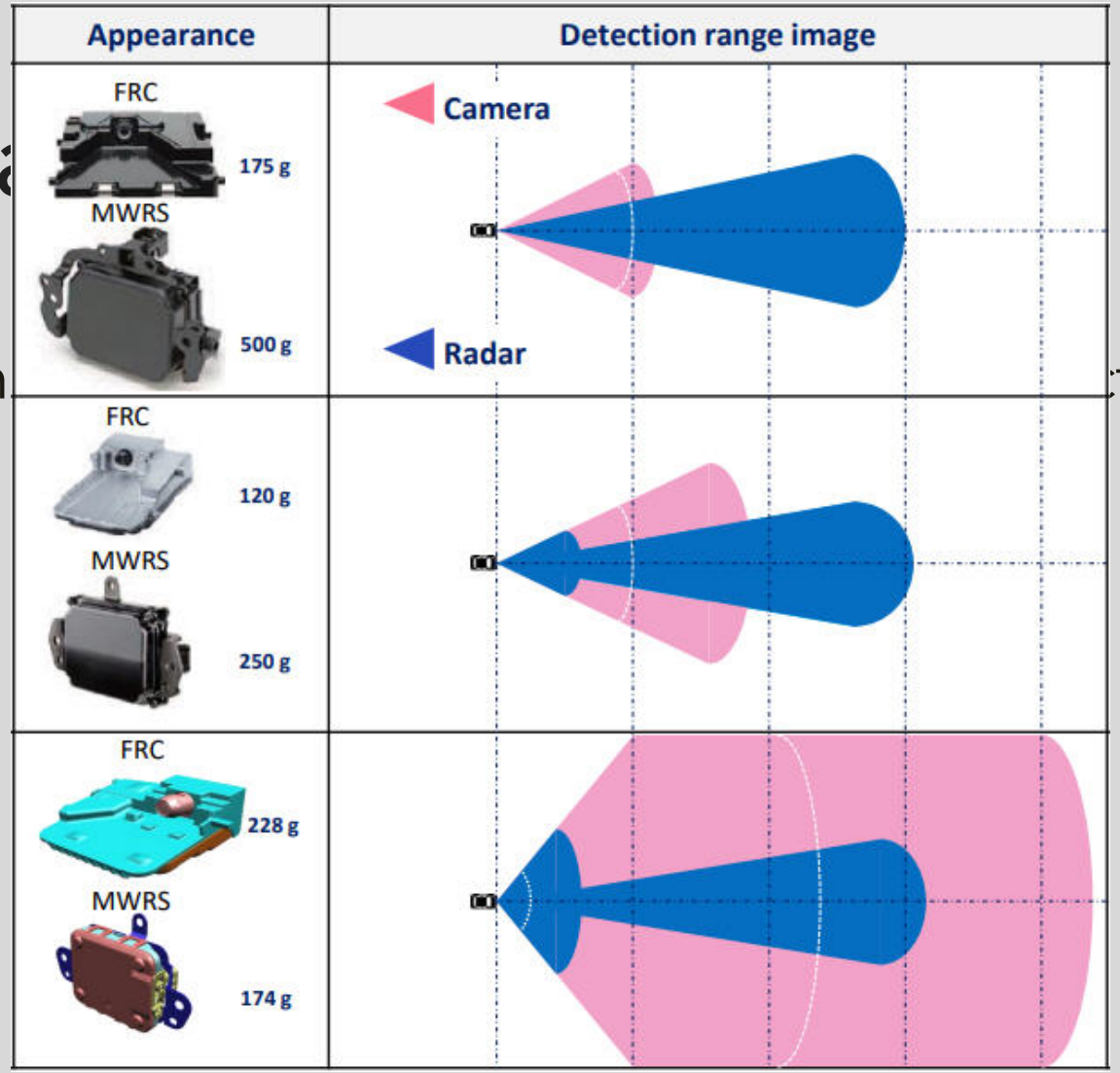
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4th Genera

- Enhanced cam tion range

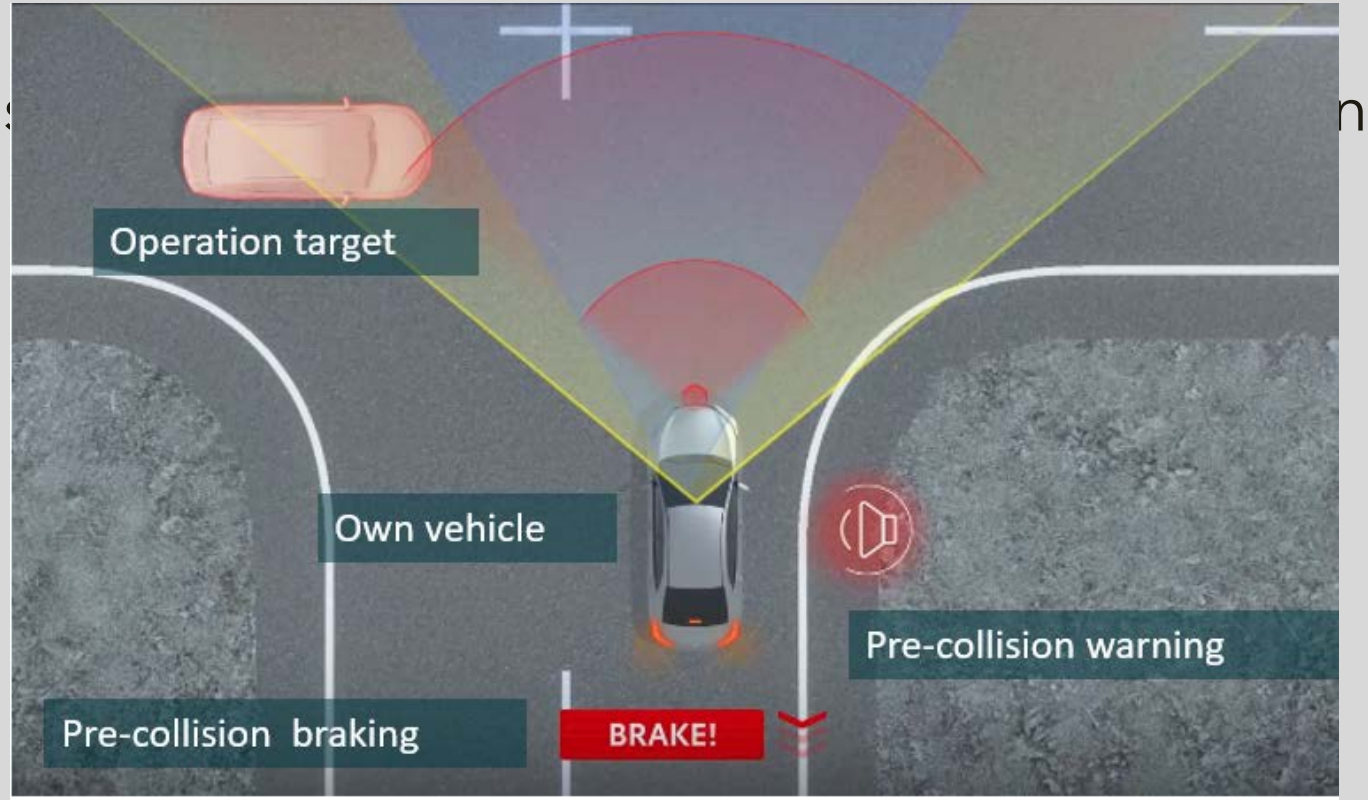


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4th Generation – TSS 3.0

- Added Cros nce support



Intersection collision avoidance support (Crossing Vehicles)

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4th Generation – TSS 3.0

Features		Toyota Safety Sense		Toyota Safety Sense 3.0	
		2.0	2.5+	Without driver monitor camera	With driver monitor camera
Operating Objects	Vehicles, Pedestrians and Bicyclist	●	●	●	●
	Oncoming Vehicle	–	●	●	●
	Motorcycle	–	–	●	●
Intersection Assistance	Right/left Turn	●	●	●	●
	Crossing Vehicle	–	–	●	●
Acceleration Suppression at Low Speed Function		–	–	●	●
Emergency Steering Assist Function		●	●	●	●

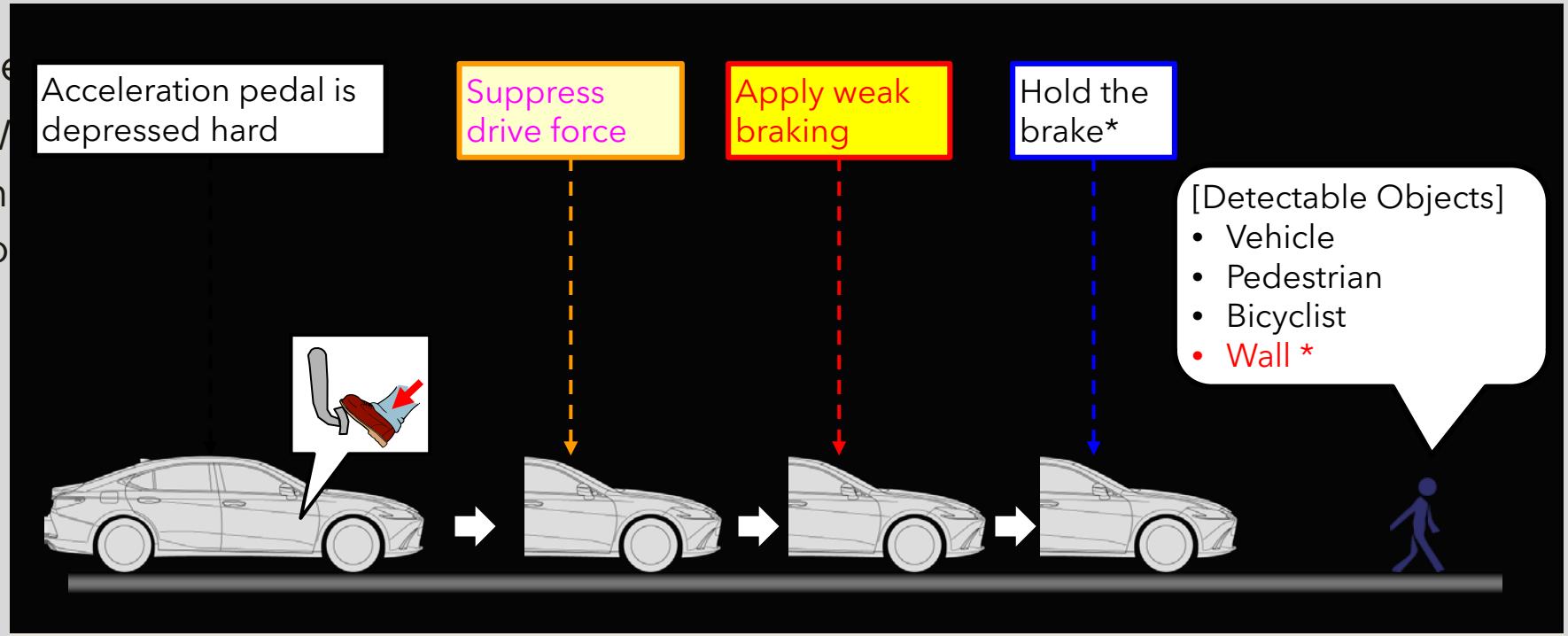
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4th Generation – TSS 3.0

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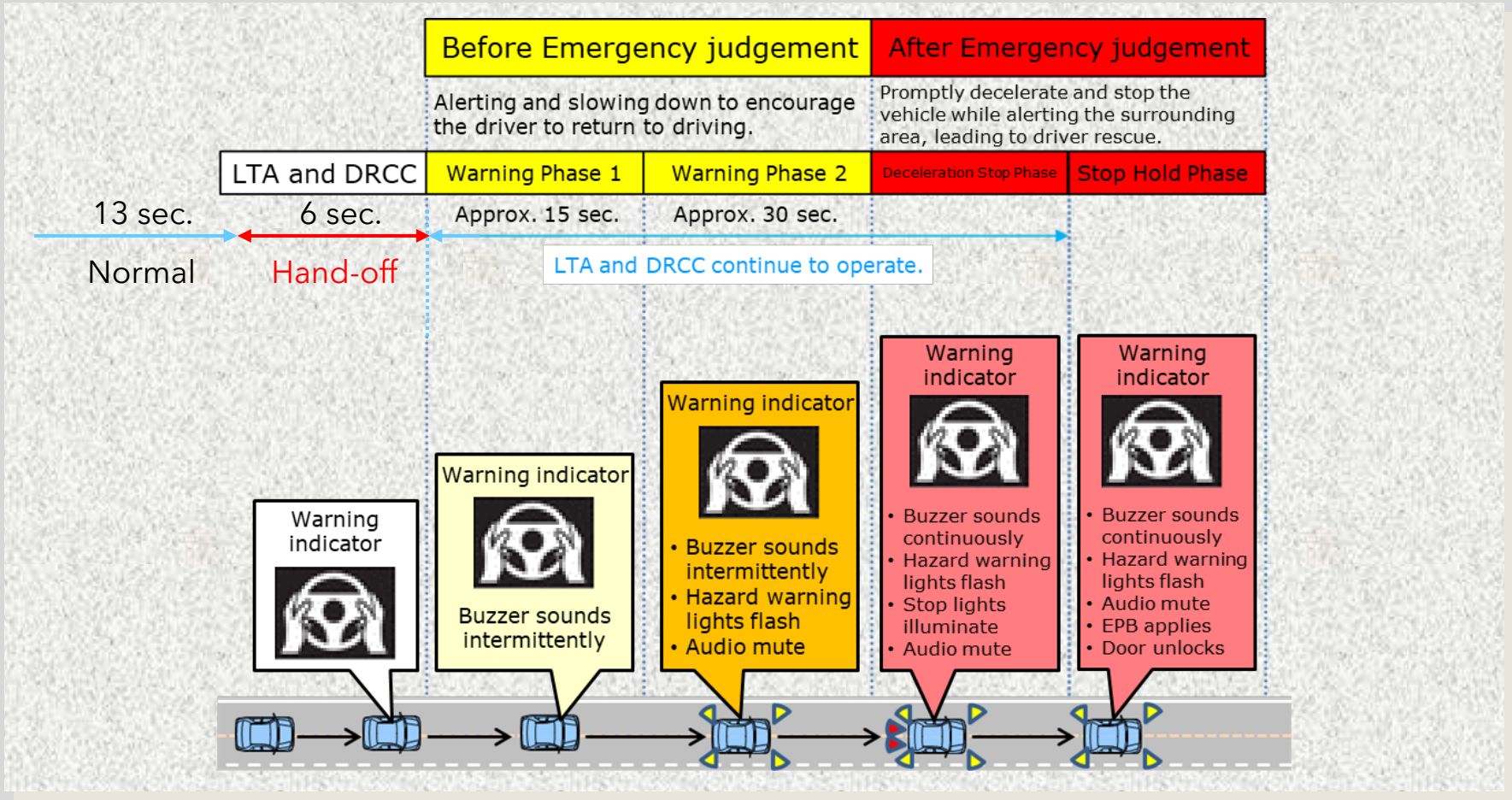
4th Generation – TSS 3.0

Emergency Driving Stop System (EDSS)

- If the driver becomes incapacitated due to sudden illness while driving, the EDSS controls the vehicle to decelerate and stop within its lane
- When using a driver monitor camera, the driver information from the camera is added to the driver abnormality determination, thus reducing the time required for determination

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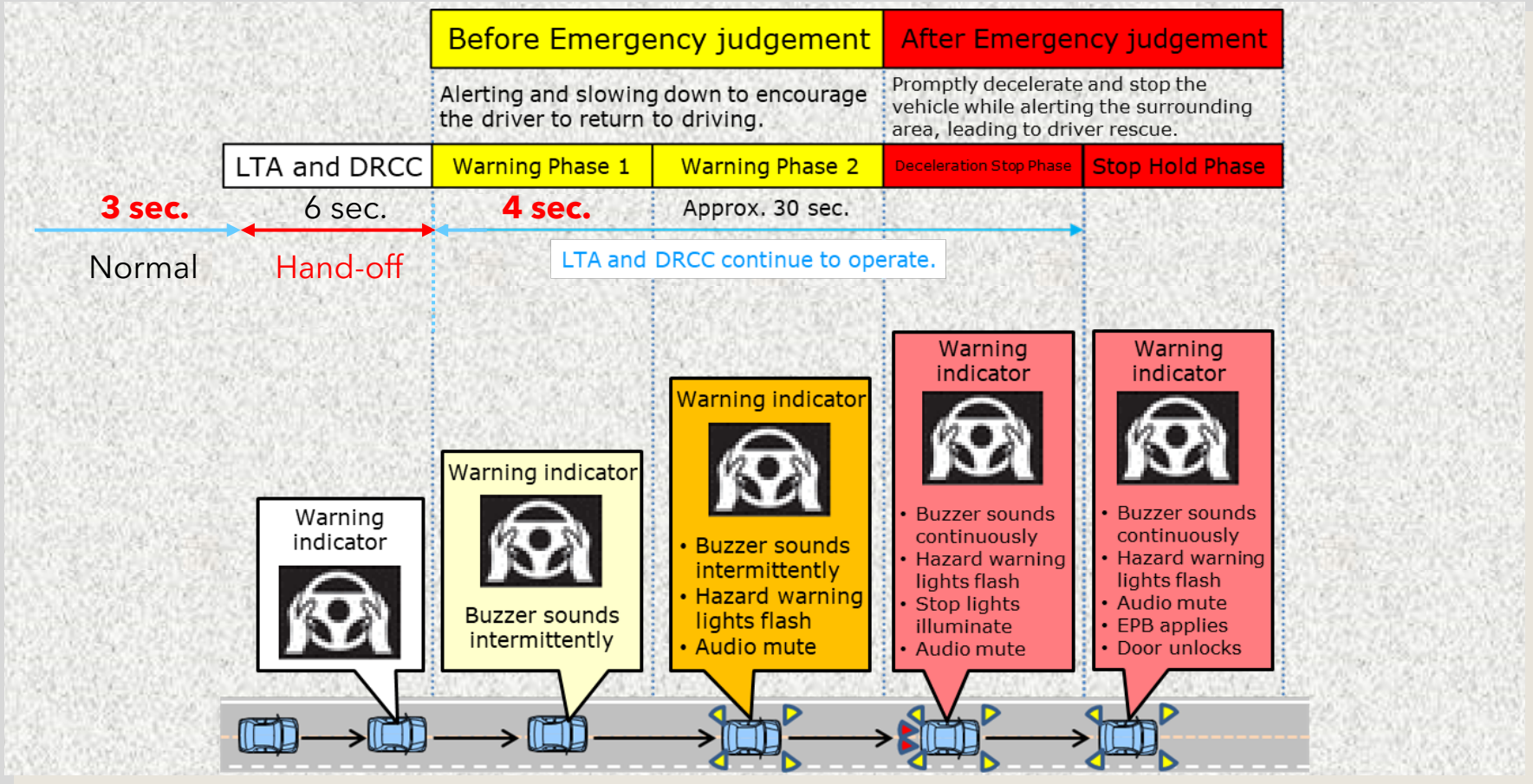
4th Generation – TSS 3.0



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4th Generation – TSS 3.0



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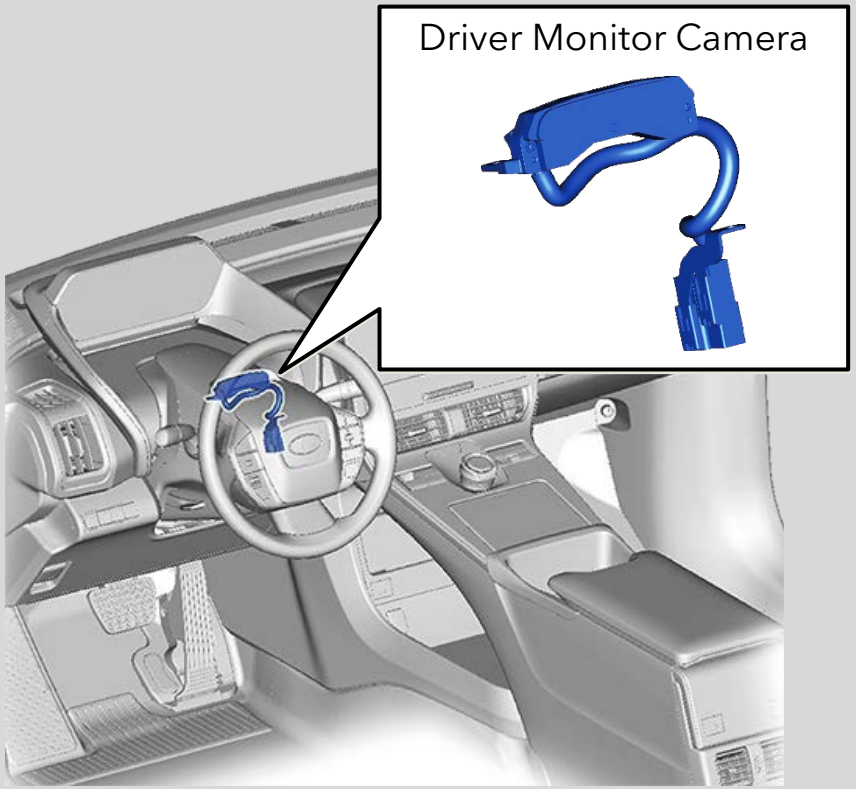
4th Generation – TSS 3.0

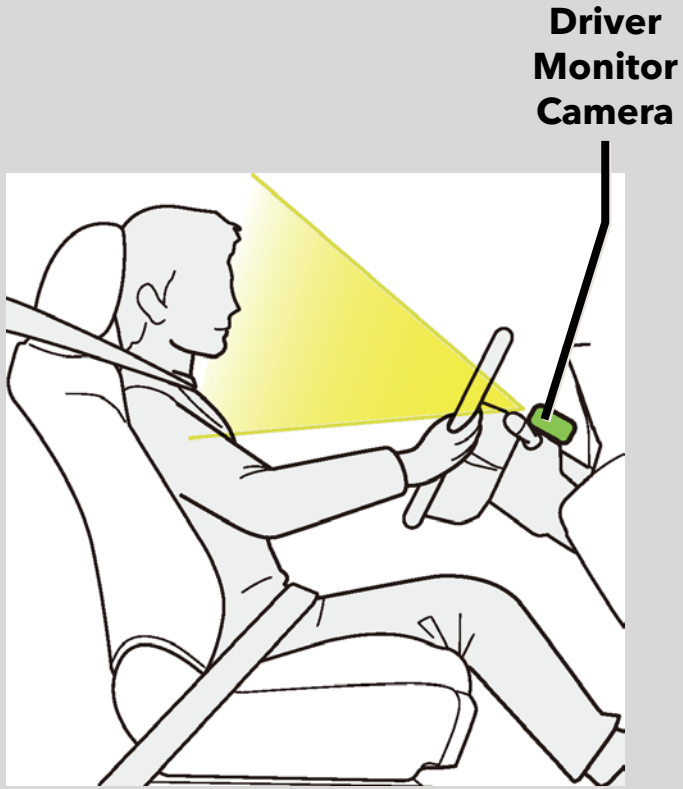
Driver Monitor Camera

- The driver monitor camera uses an infrared camera in order to detect:
  - The direction the driver is facing
  - The driver's line of sight
  - Whether the driver's eyes are open or closed

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4th Generation – TSS 3.0





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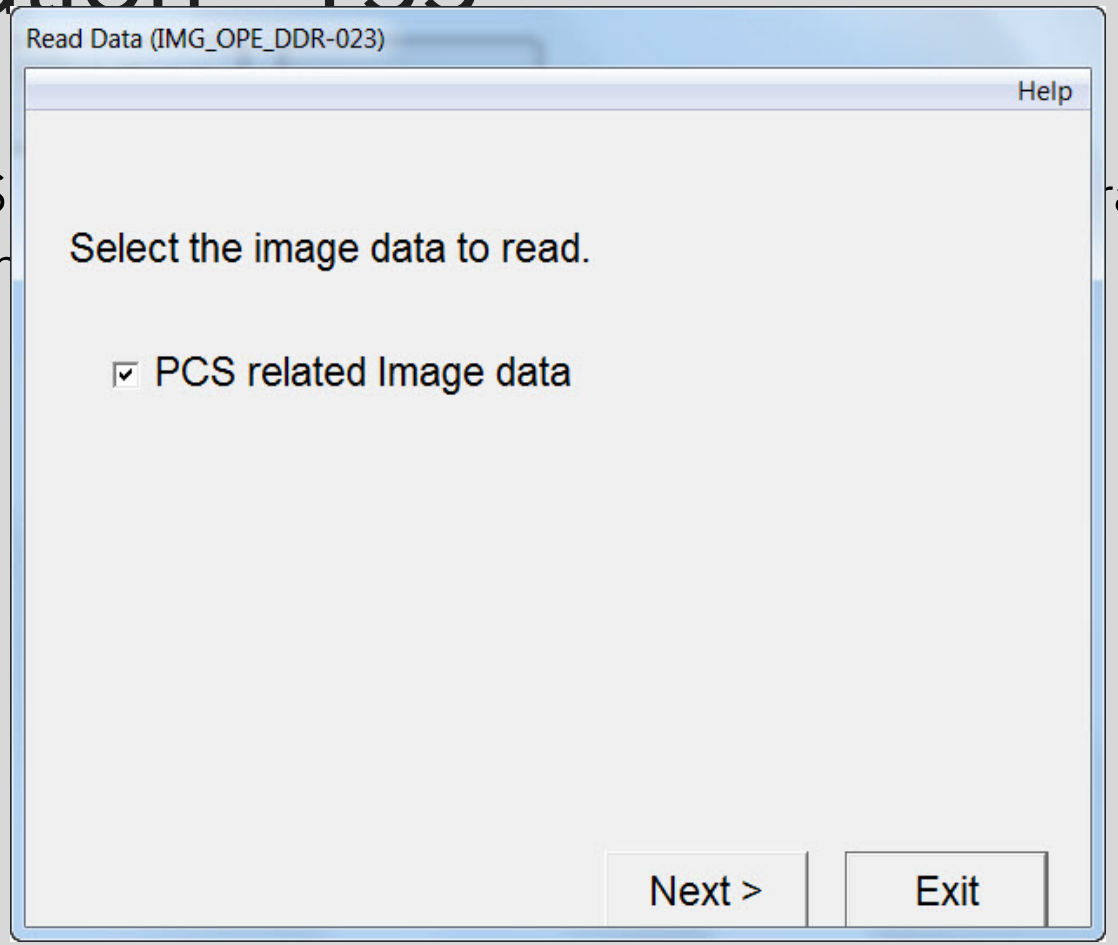
Accessing Data with Toyota Techstream

- Vehicle Control History (VCH) data is stored in the Airbag ECU (EEPROM)
- The images are stored in the Front Recognition Camera module (EEPROM)
- VCH / FFD data is the Vehicle Control History (VCH) data and the image data linked together for certain event thresholds
- The threshold to record image data is dependent on the generation of TSS

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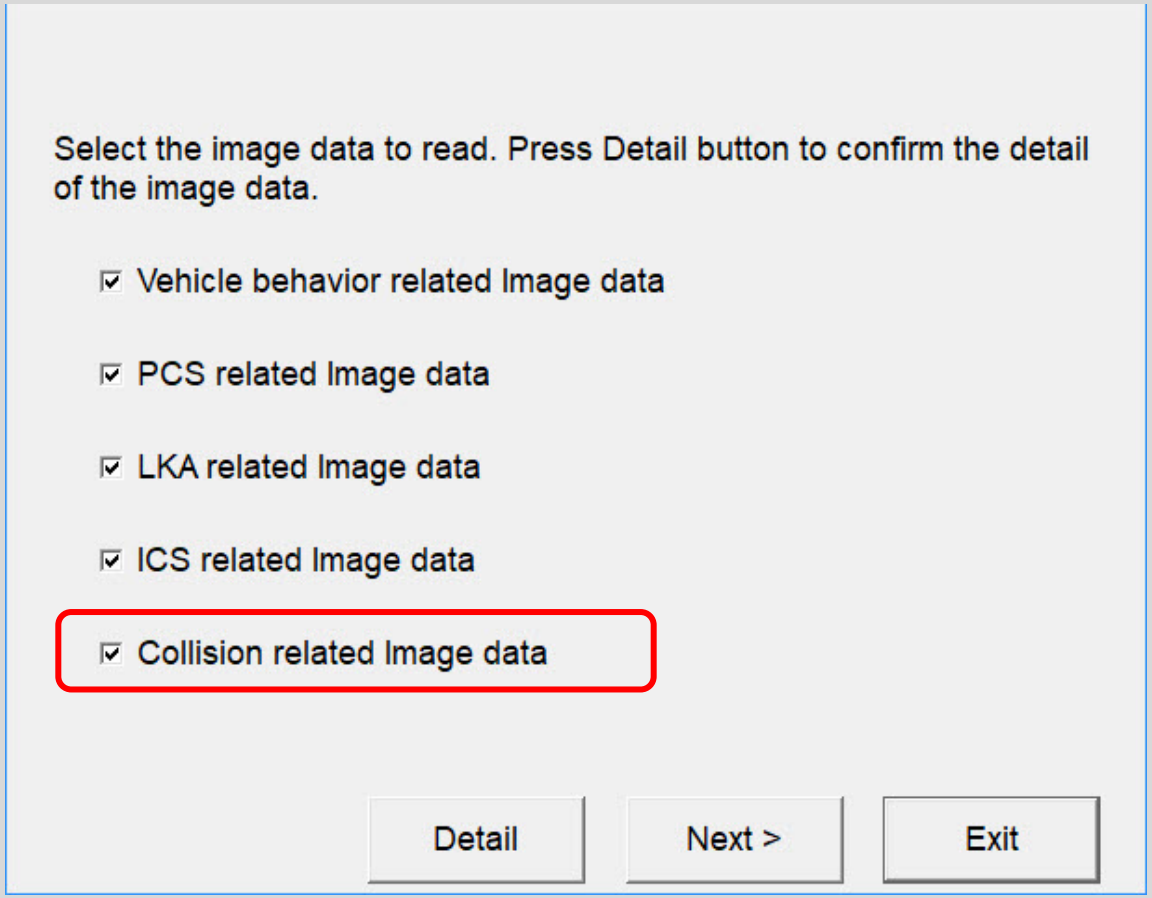
1st Generation – TSS

- 1st Generation TSS raking (PB/FPB) Operation or wher



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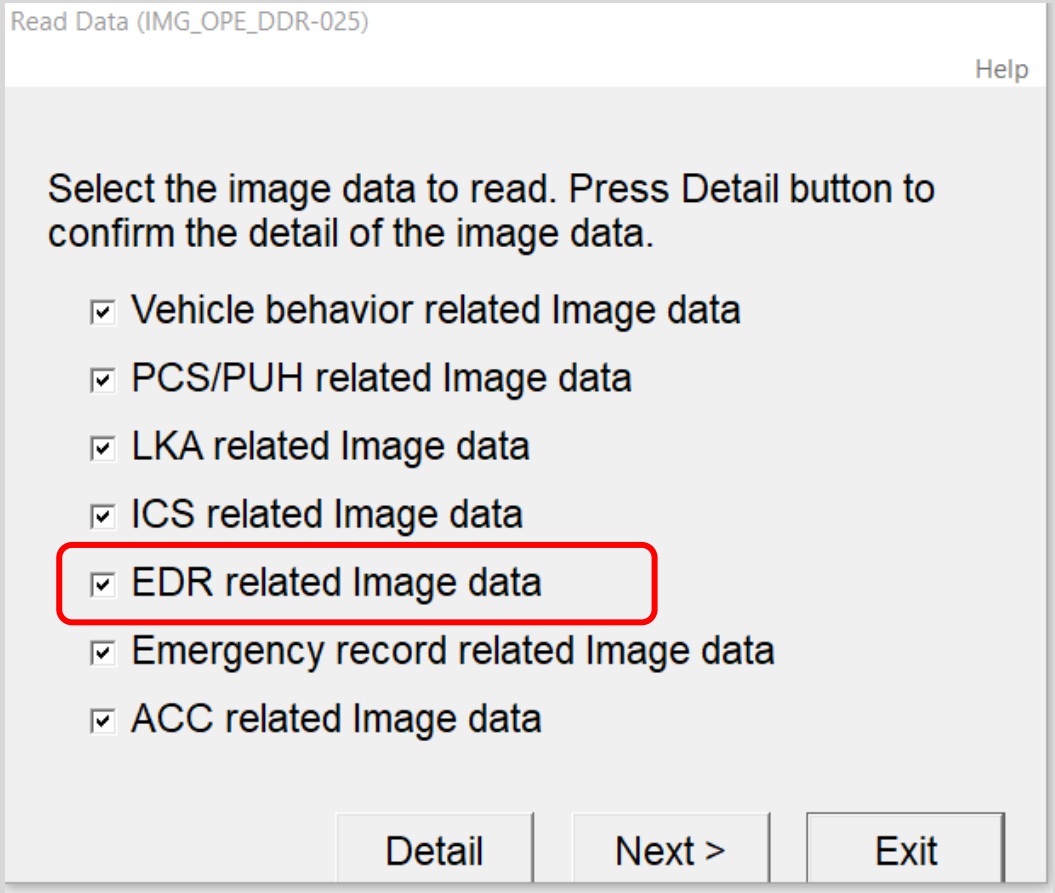
2nd Generation – TSS 2.0



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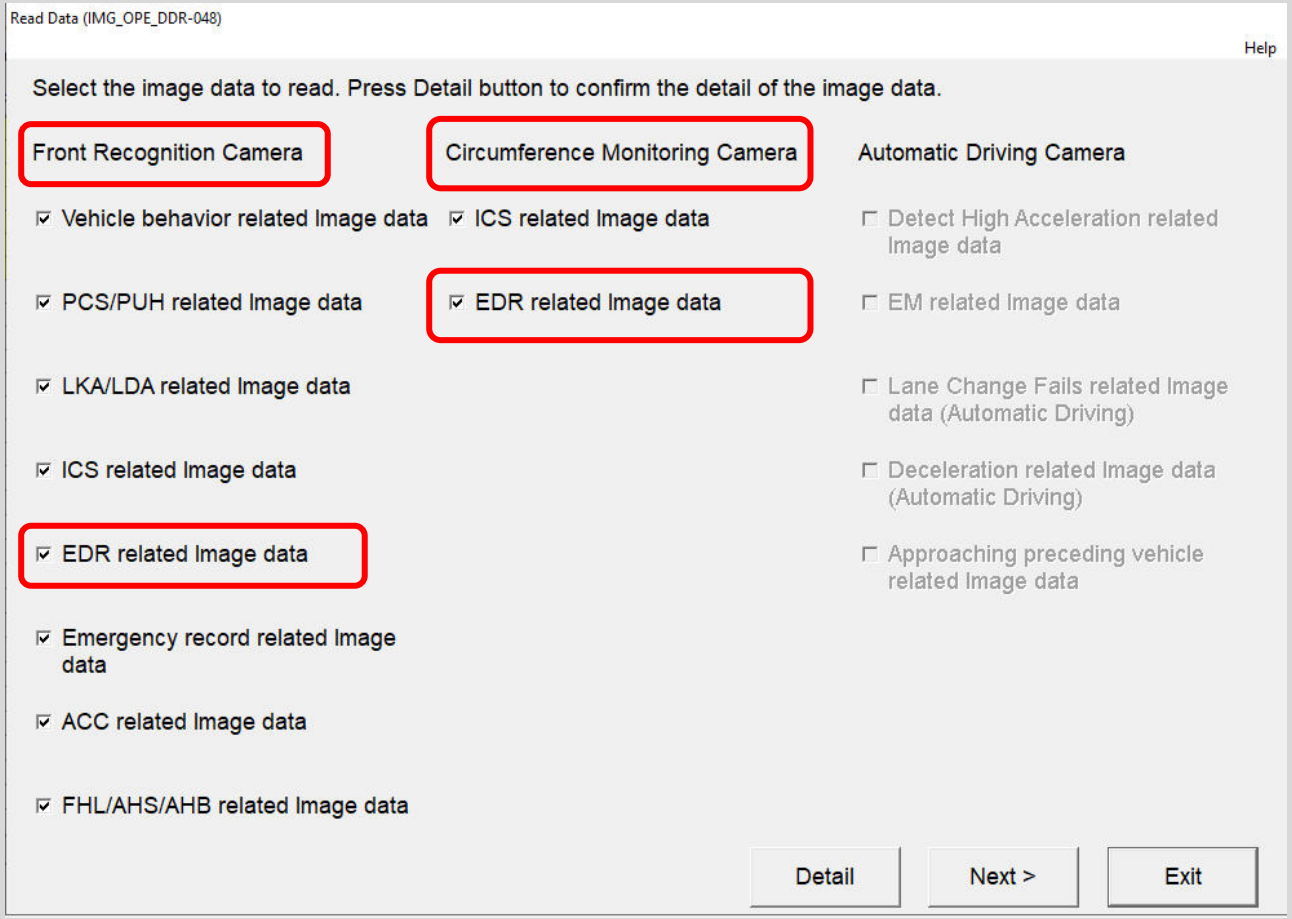
3rd Generation – TSS 2.5 & 2.5+



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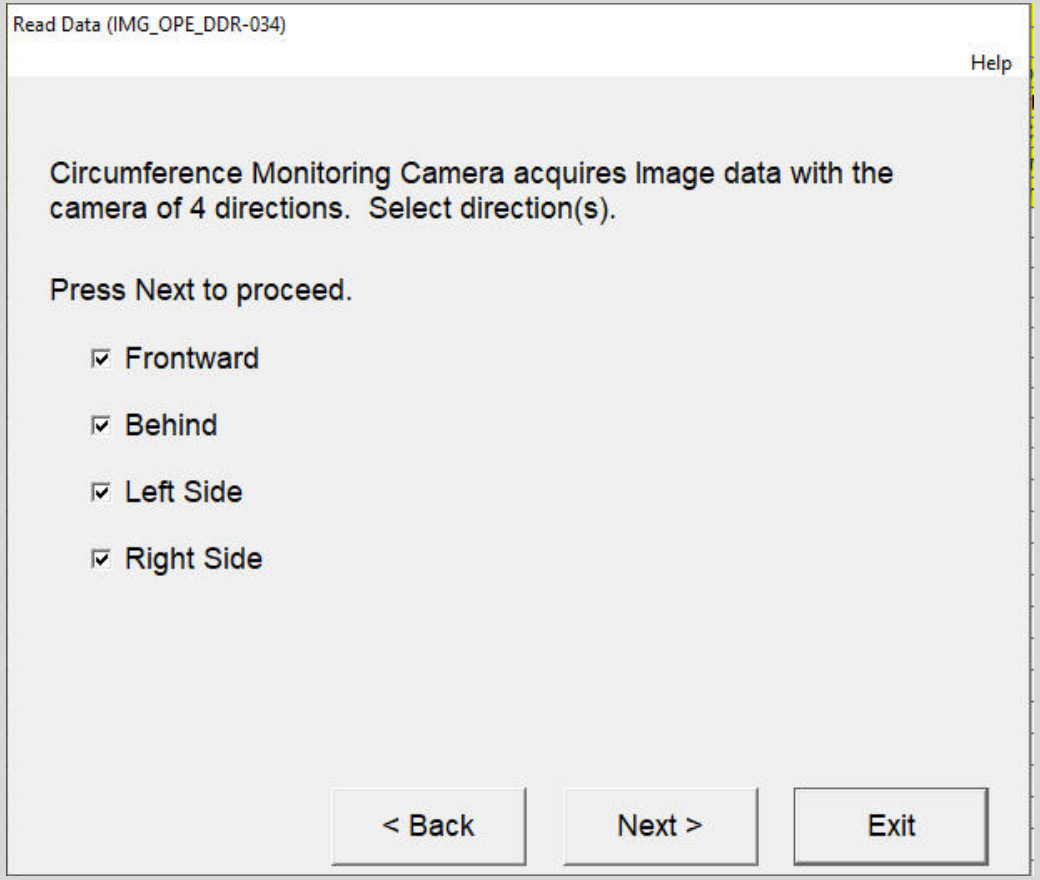
4th Generation – TSS 3.0



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4th Generation – TSS 3.0



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Accessing Data with Toyota Techstream

Commercially available Toyota Techstream software

http://techinfo.toyota.com

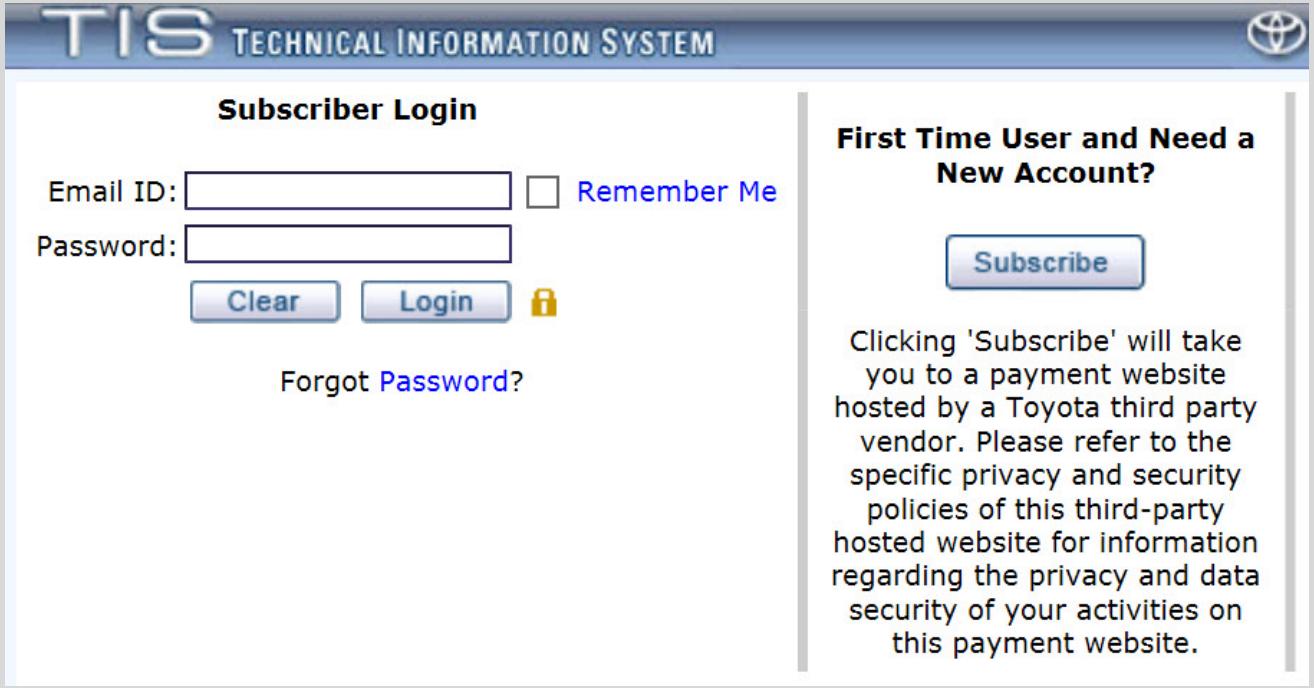


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Accessing Data with Toyota Techstream

- Need to register online and create a profile to Subscribe



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Accessing Data with Toyota Techstream

- Purchase software subscription https://techinfo.snapon.com/TIS/Register.aspx

Subscription Type	Standard	Professional Diagnostic	Security Professional*
2 Day	**$20**	**$65**	**$65**
Monthly	**$90**	**NA**	**NA**
Yearly	**$480**	**$1295**	**$1295**

NOTE: All subscriptions expire at midnight, Pacific Time on the last day of your term.
This means a two-day subscription may actually provide greater than 48 hours of usable access.

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Accessing Data with Toyota Techstream

Toyota Techstream LITE kit

- Laptop Computer
- Toyota Techstream Lite software
- Mongoose Cable (or CDR 900 Cable) to connect through the DLC



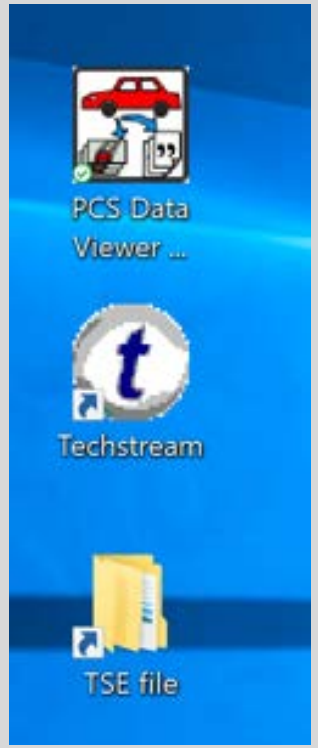


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Accessing Data with Toyota Techstream

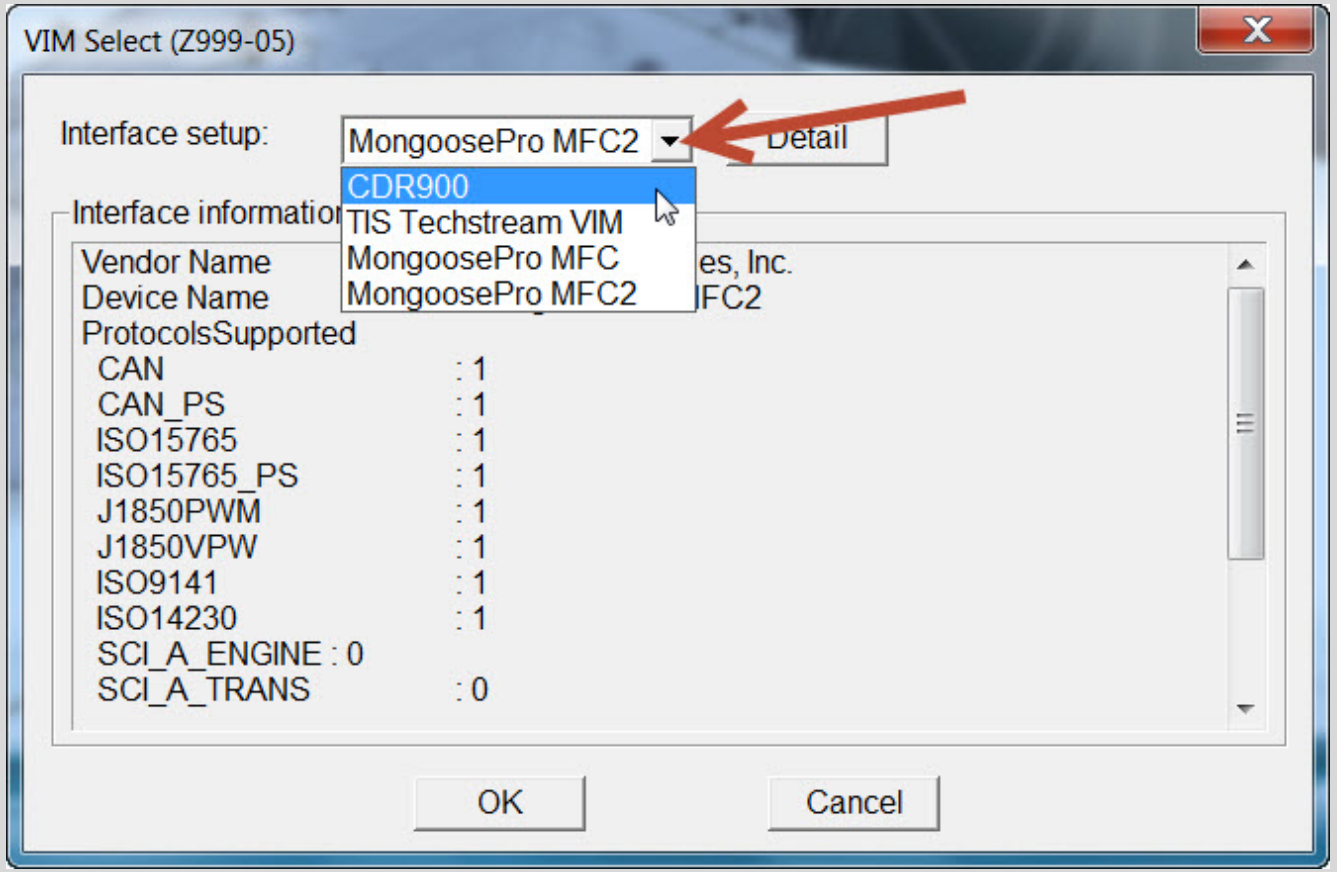
Following the setup of the Techstream you will find the following Icons on your desktop:

- PCS Data Viewer – launches the tool to view the VDAS file
- TechStream – launches the Techstream application
- TSE File – directory for information stored using TechStream



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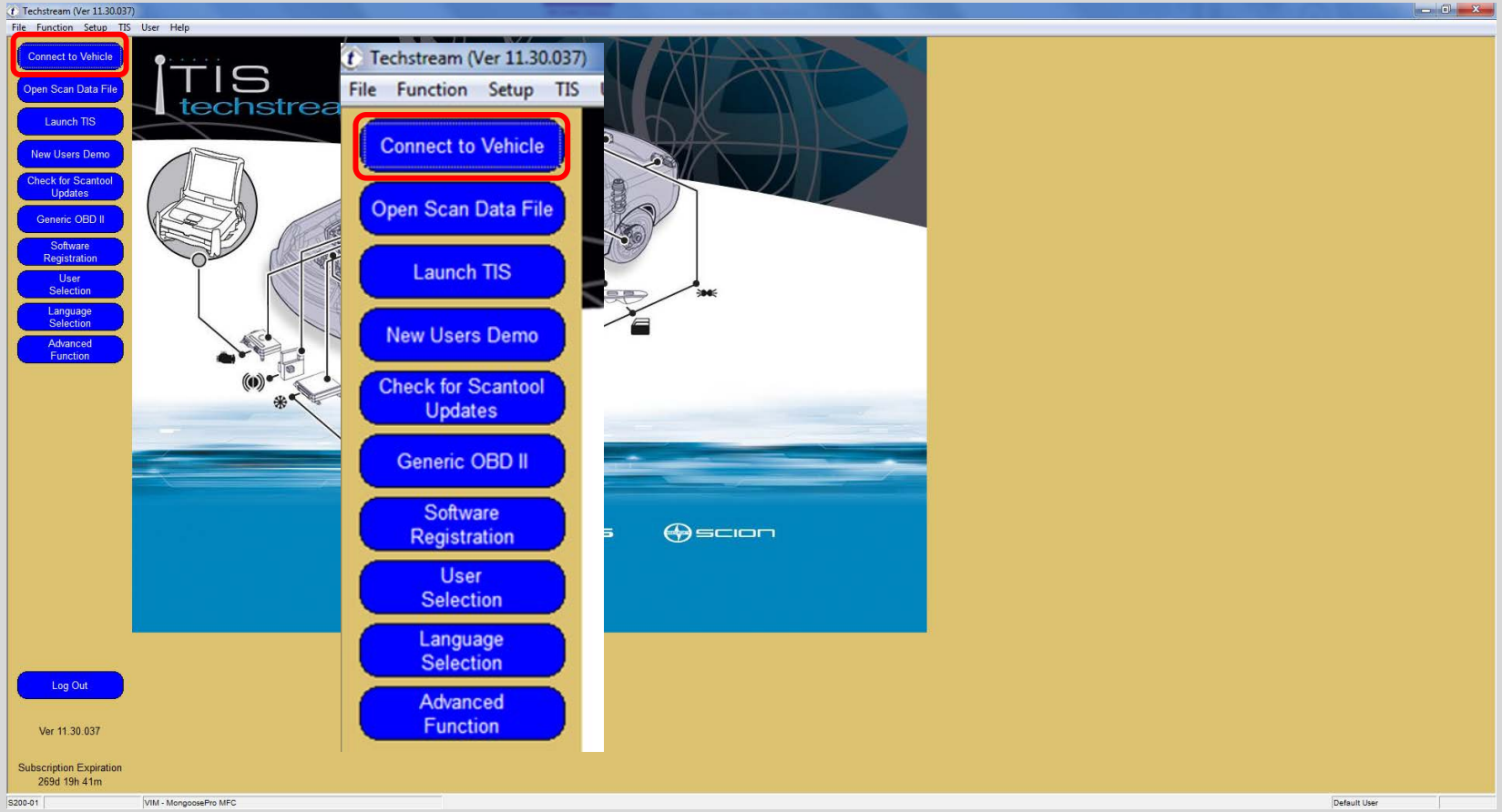
Accessing Data with Toyota Techstream



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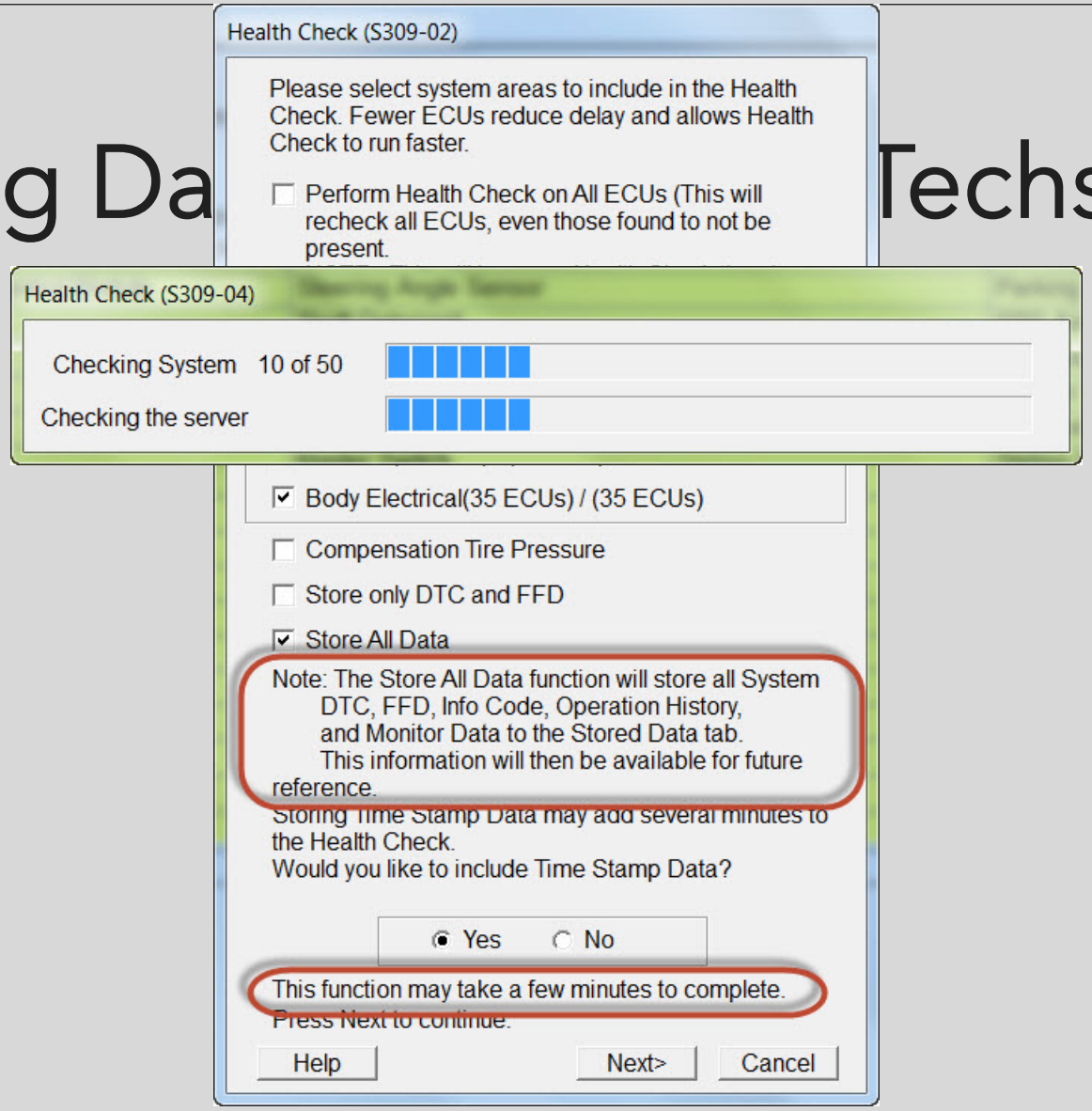
Accessing Data with Toyota Techstream



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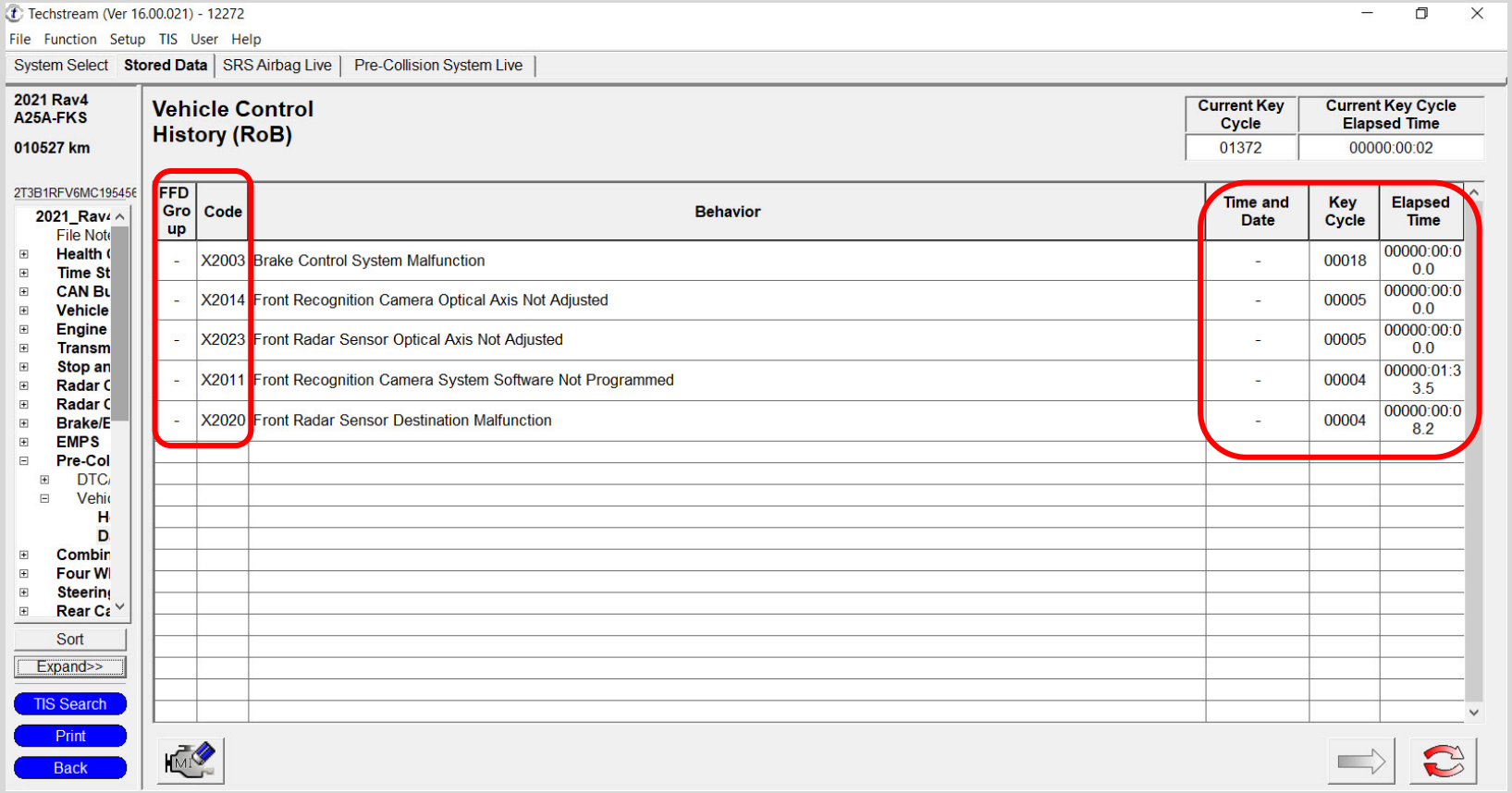
Accessing Da Techstream



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Readout of Behavior (RoB)



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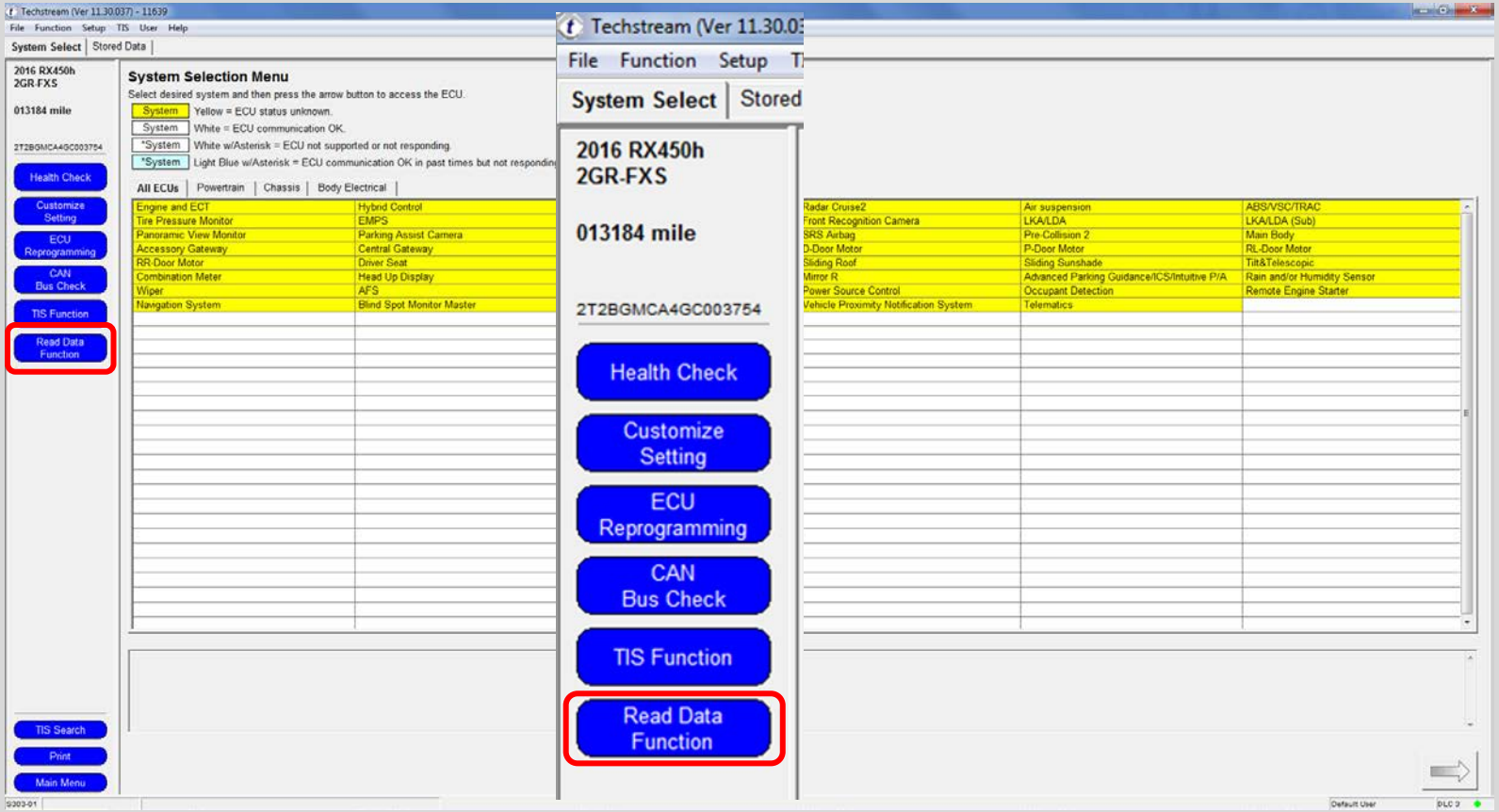
Readout of Behavior (RoB)

CHECK VEHICLE CONTROL HISTORY (FRONT RADAR SENSOR SYSTEM)

Pre-Collision System Unavailable Clean Sensor	X2026	Front Radar Sensor Dirt	History of system operation being temporarily suspended due to millimeter wave radar sensor assembly not being able to detect objects.	Check if any of the following conditions are met: • A non-genuine front grille emblem is installed • The millimeter wave radar sensor assembly or front grille emblem and its surrounding area are dirty • An accessory, such as bull bars, a grille guard, etc., which interferes with millimeter radio waves is installed • The front grille emblem or its surrounding area has been painted or has a sticker attached • There is foreign matter attached to the front grille emblem or millimeter wave radar sensor assembly **Cause:** The millimeter wave radar sensor assembly is dirty or obstructed by foreign matter.
Unavailable		Optical Axis Misalignment)	axis misalignment (vertical).	

•• PROTECTED 関係者外秘 PROTÉGÉ

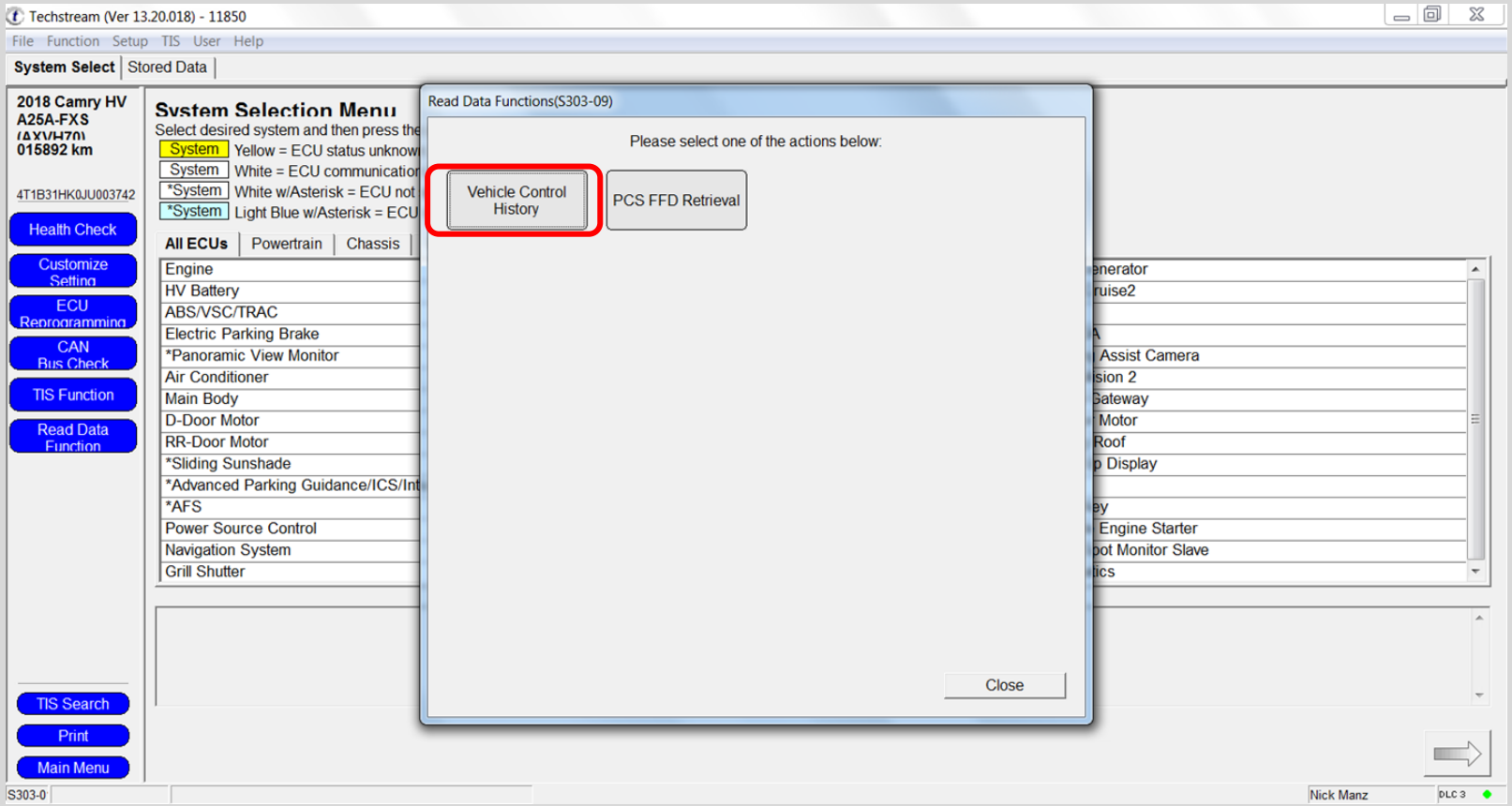
Accessing Data with Toyota Techstream



WREX 2023 – April 17-21, 2023

•• PROTECTED 関係者外秘 PROTÉGÉ

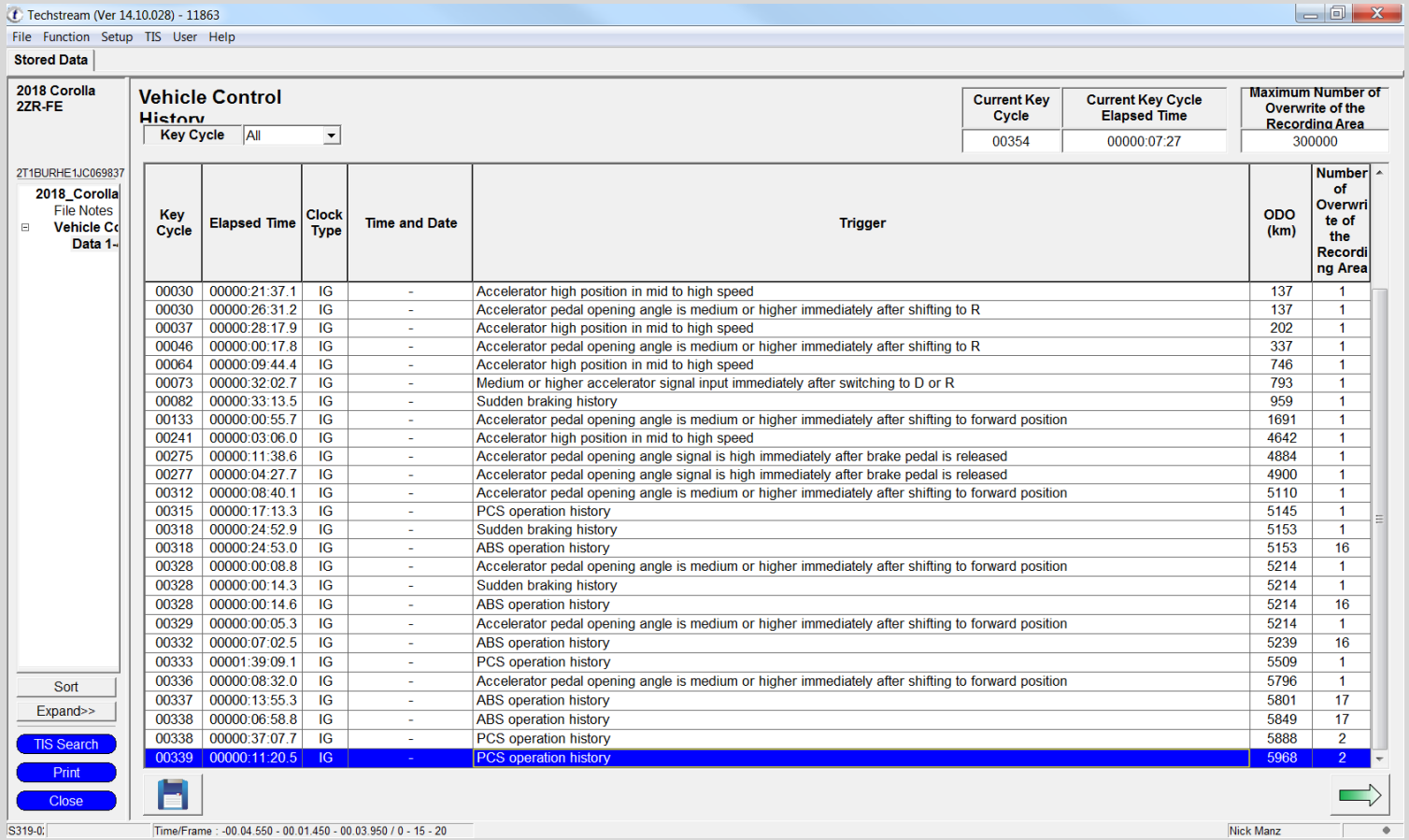
Accessing Data with Toyota Techstream



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Vehicle Control History (VCH) Data



Techstream (Ver 14.10.028) - 11863

File Function Setup TIS User Help

Stored Data

2018 Corolla
2ZR-FE

2T1BURHE1JC069837

2018_Corolla
File Notes
Vehicle Co
Data 1~

Sort
Expand>>
TIS Search
Print
Close

Vehicle Control History

Key Cycle: All

Current Key Cycle	Current Key Cycle Elapsed Time	Maximum Number of Overwrite of the Recording Area
00354	00000:07:27	300000

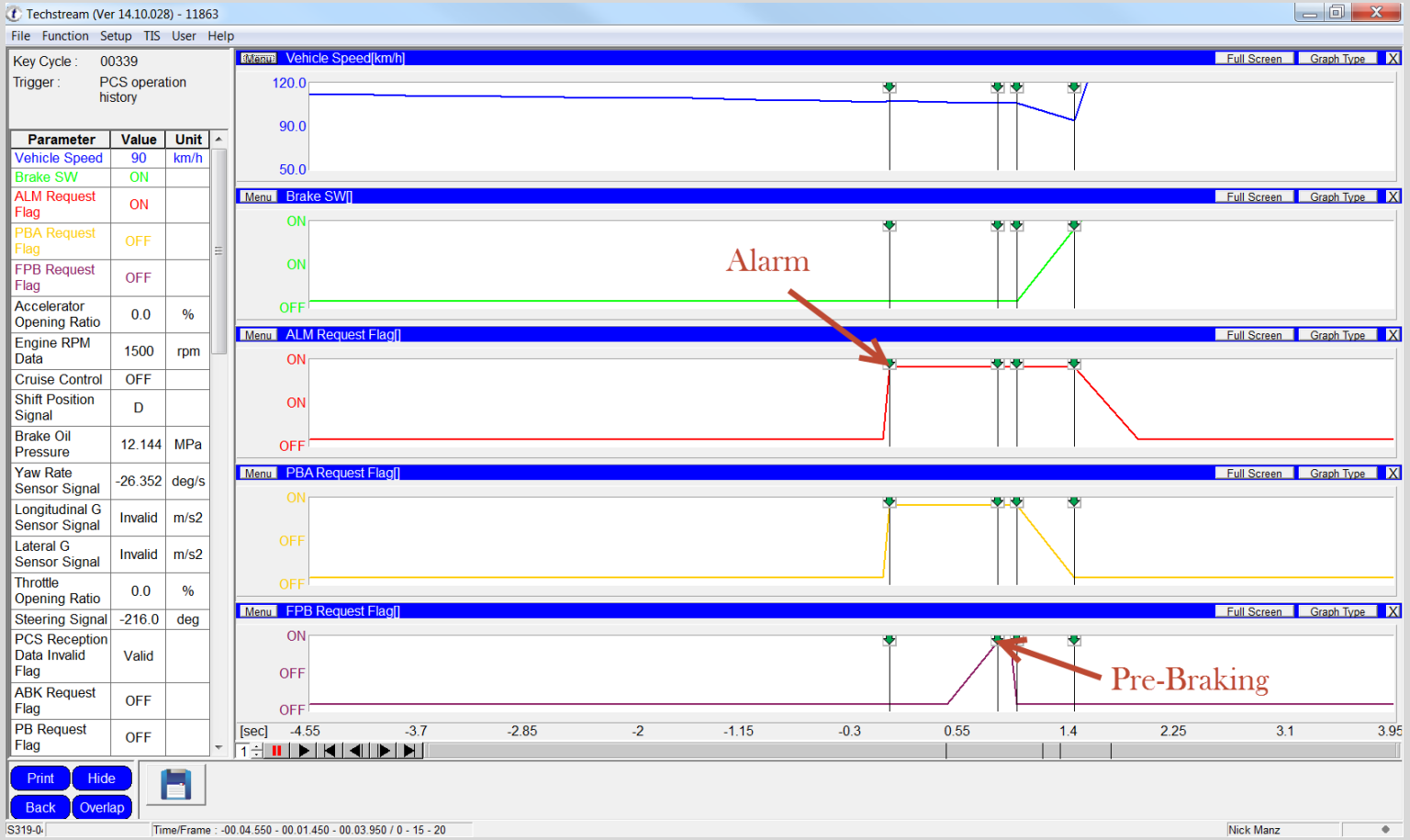
Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO (km)	Number of Overwrite of the Recording Area
00030	00000:21:37.1	IG	-	Accelerator high position in mid to high speed	137	1
00030	00000:26:31.2	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	137	1
00037	00000:28:17.9	IG	-	Accelerator high position in mid to high speed	202	1
00046	00000:00:17.8	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	337	1
00064	00000:09:44.4	IG	-	Accelerator high position in mid to high speed	746	1
00073	00000:32:02.7	IG	-	Medium or higher accelerator signal input immediately after switching to D or R	793	1
00082	00000:33:13.5	IG	-	Sudden braking history	959	1
00133	00000:00:55.7	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	1691	1
00241	00000:03:06.0	IG	-	Accelerator high position in mid to high speed	4642	1
00275	00000:11:38.6	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	4884	1
00277	00000:04:27.7	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	4900	1
00312	00000:08:40.1	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	5110	1
00315	00000:17:13.3	IG	-	PCS operation history	5145	1
00318	00000:24:52.9	IG	-	Sudden braking history	5153	1
00318	00000:24:53.0	IG	-	ABS operation history	5153	16
00328	00000:00:08.8	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	5214	1
00328	00000:00:14.3	IG	-	Sudden braking history	5214	1
00328	00000:00:14.6	IG	-	ABS operation history	5214	16
00329	00000:00:05.3	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	5214	1
00332	00000:07:02.5	IG	-	ABS operation history	5239	16
00333	00001:39:09.1	IG	-	PCS operation history	5509	1
00336	00000:08:32.0	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	5796	1
00337	00000:13:55.3	IG	-	ABS operation history	5801	17
00338	00000:06:58.8	IG	-	ABS operation history	5849	17
00338	00000:37:07.7	IG	-	PCS operation history	5888	2
00339	00000:11:20.5	IG	-	PCS operation history	5968	2

S319-0: Time/Frame : -00.04.550 - 00.01.450 - 00.03.950 / 0 - 15 - 20 Nick Manz

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•• PROTECTED 関係者外秘 PROTÉGÉ

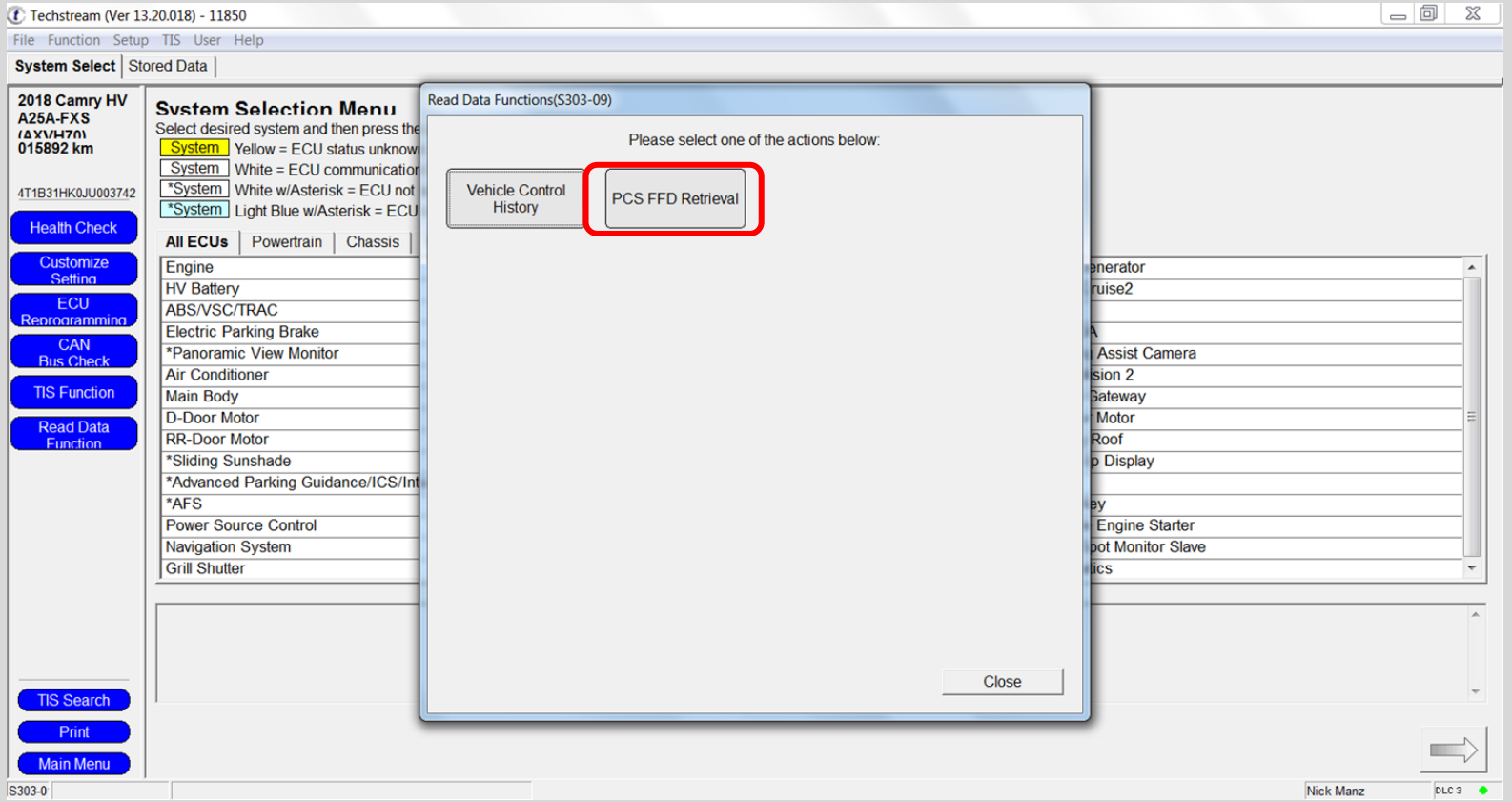
Vehicle Control History (VCH) Data



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•• PROTECTED 関係者外秘 PROTÉGÉ

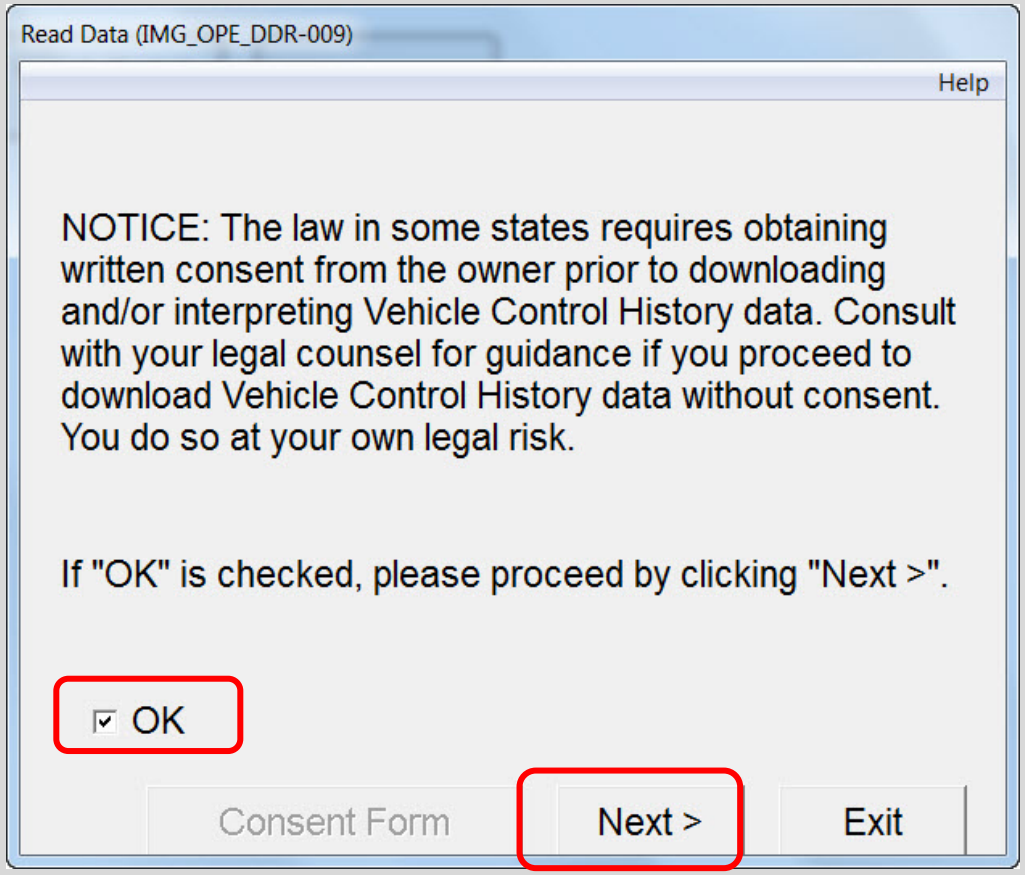
Accessing Data with Toyota Techstream



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•• PROTECTED 関係者外秘 PROTÉGÉ

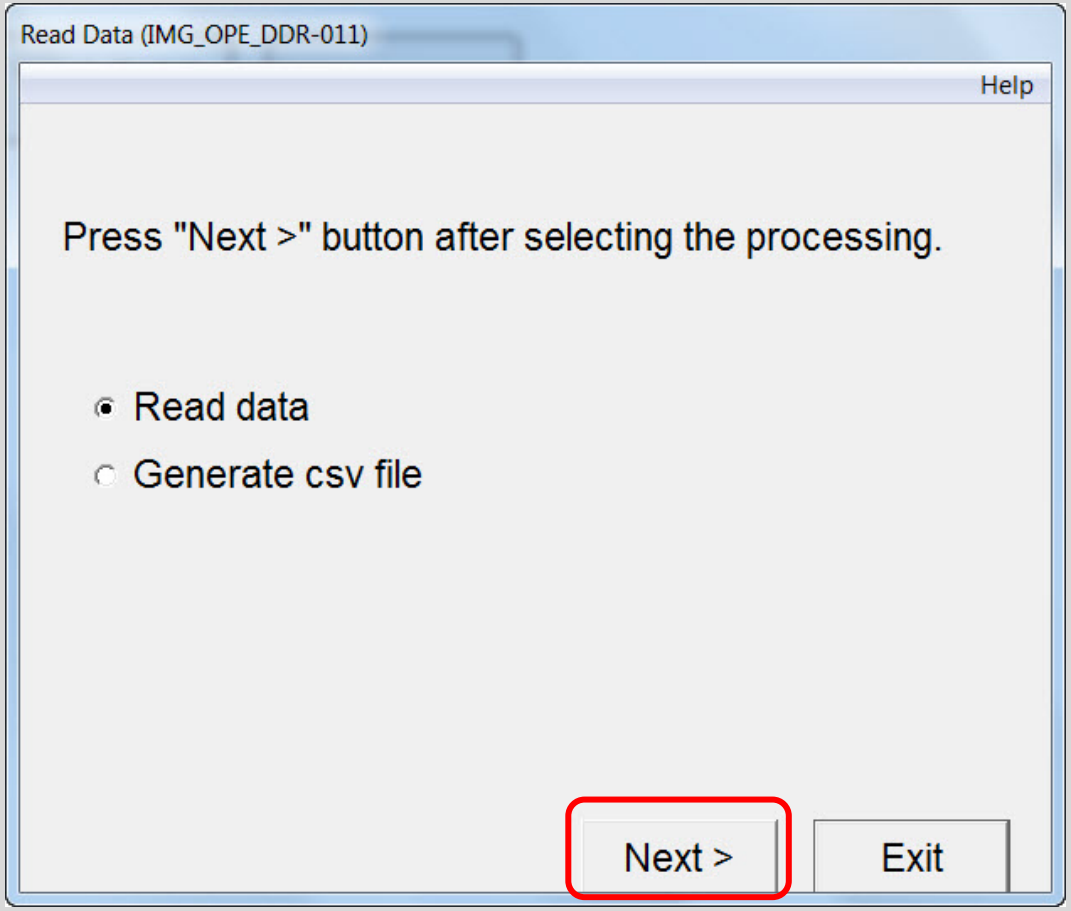
Accessing Data with Toyota Techstream



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•• PROTECTED 関係者外秘 PROTÉGÉ

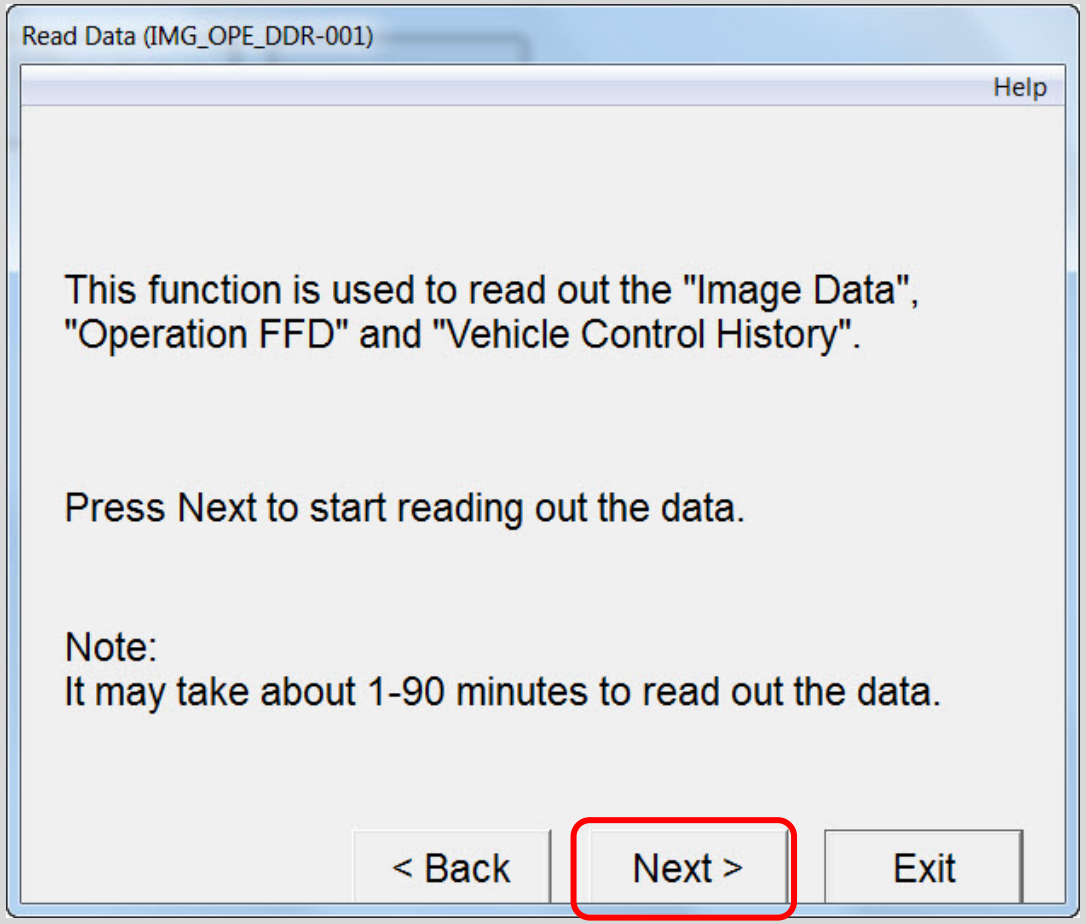
Accessing Data with Toyota Techstream



WREX 2023 – April 17-21, 2023

•• PROTECTED 関係者外秘 PROTÉGÉ

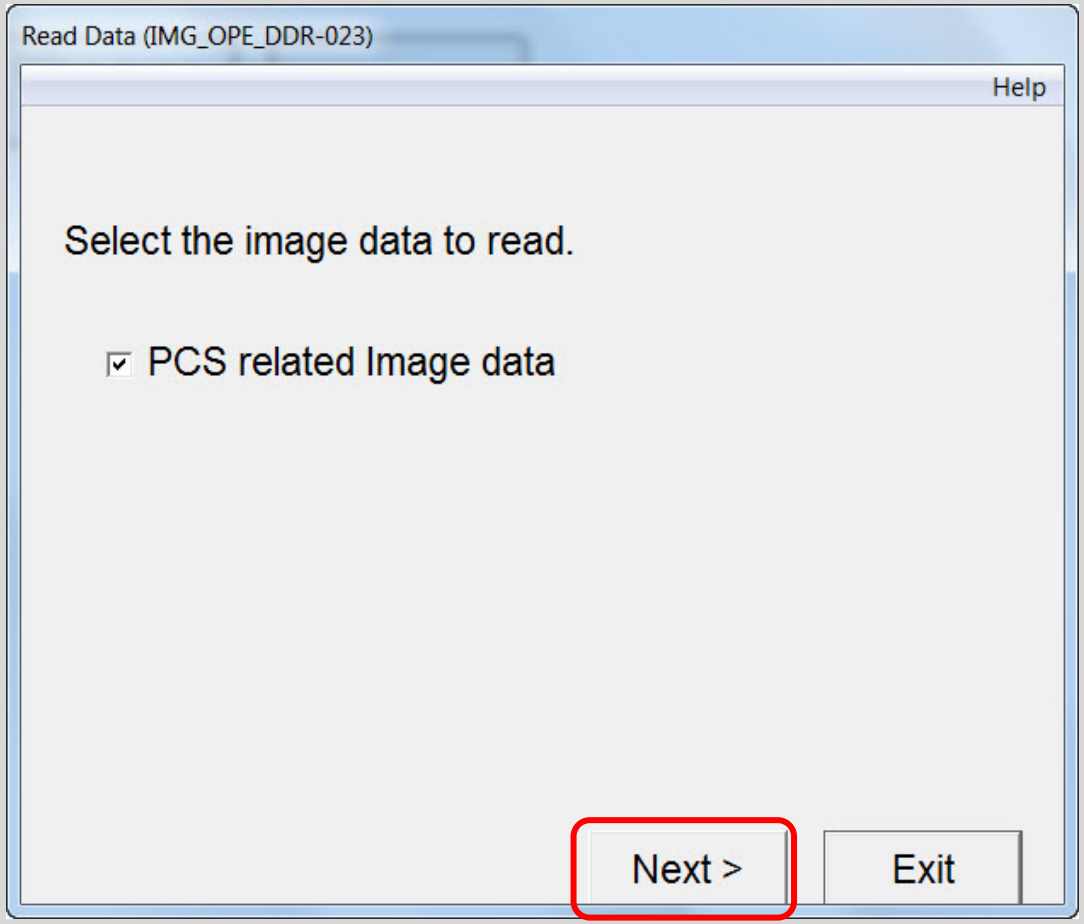
Accessing Data with Toyota Techstream



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•• PROTECTED 関係者外秘 PROTÉGÉ

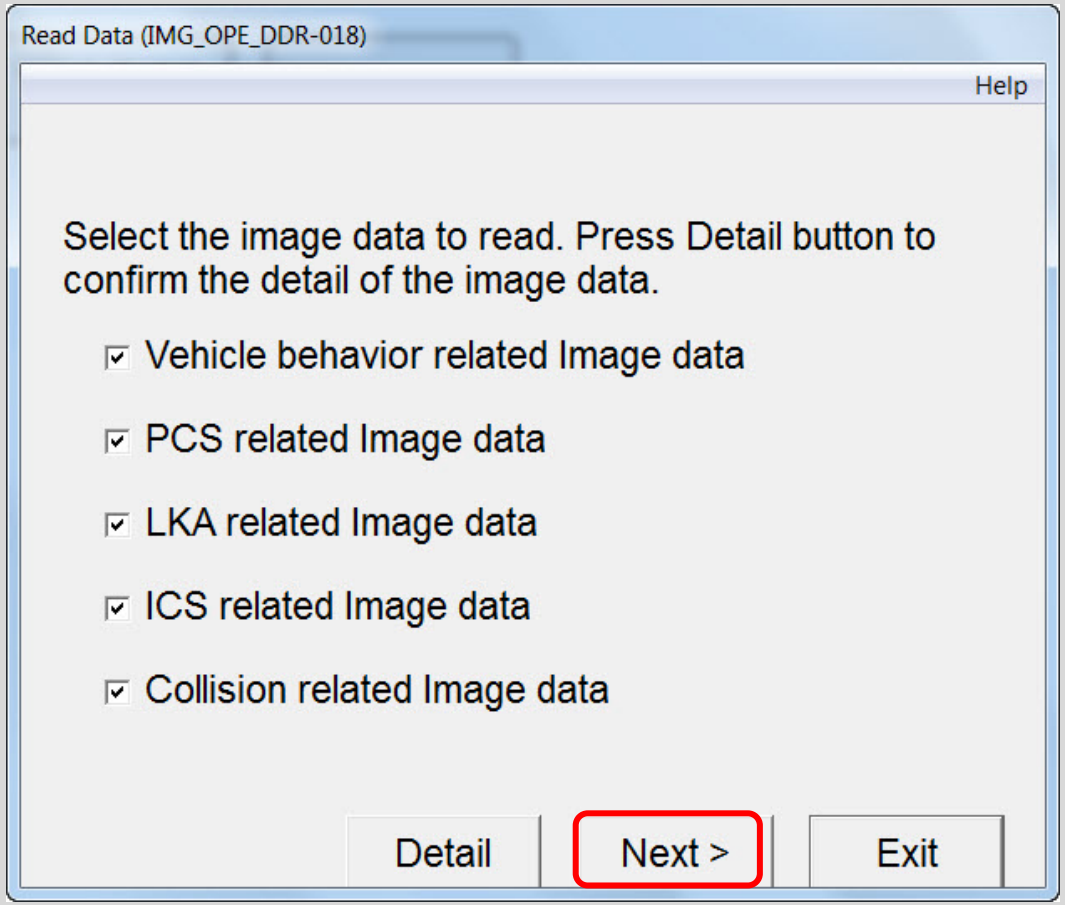
1st Generation TSS



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•• PROTECTED 関係者外秘 PROTÉGÉ

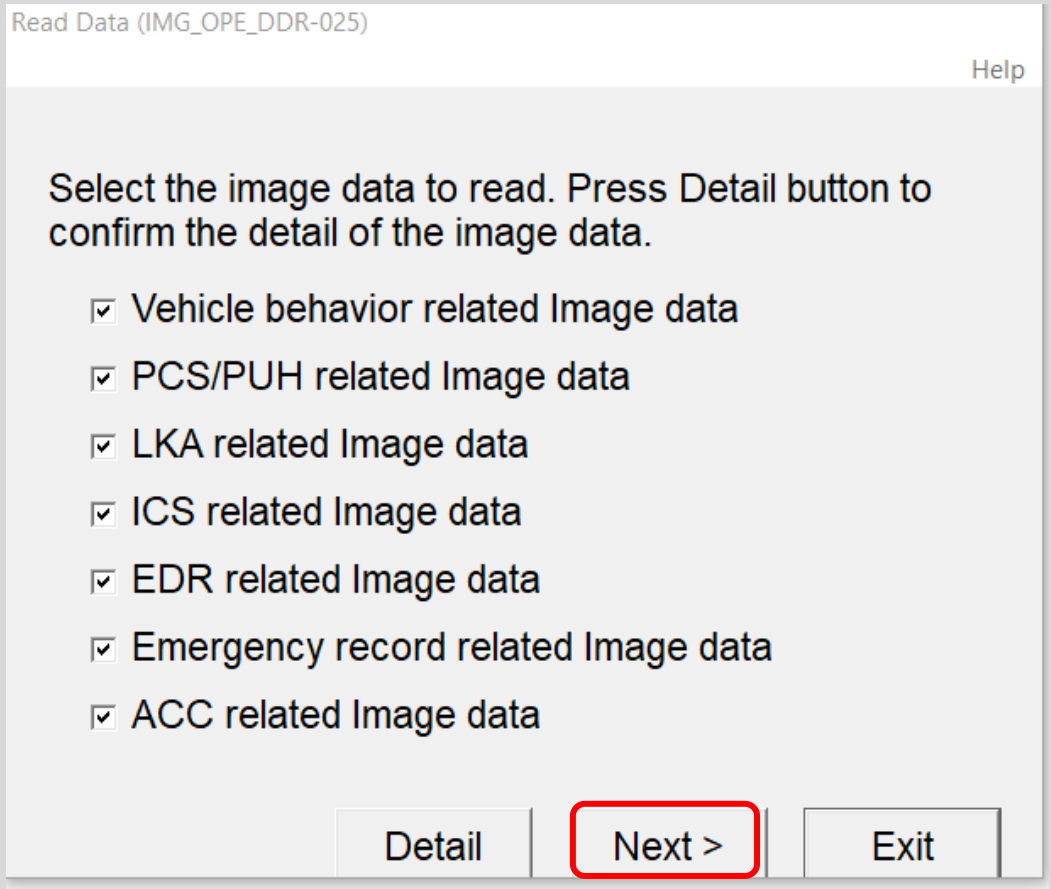
2nd Generation TSS



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•• PROTECTED 関係者外秘 PROTÉGÉ

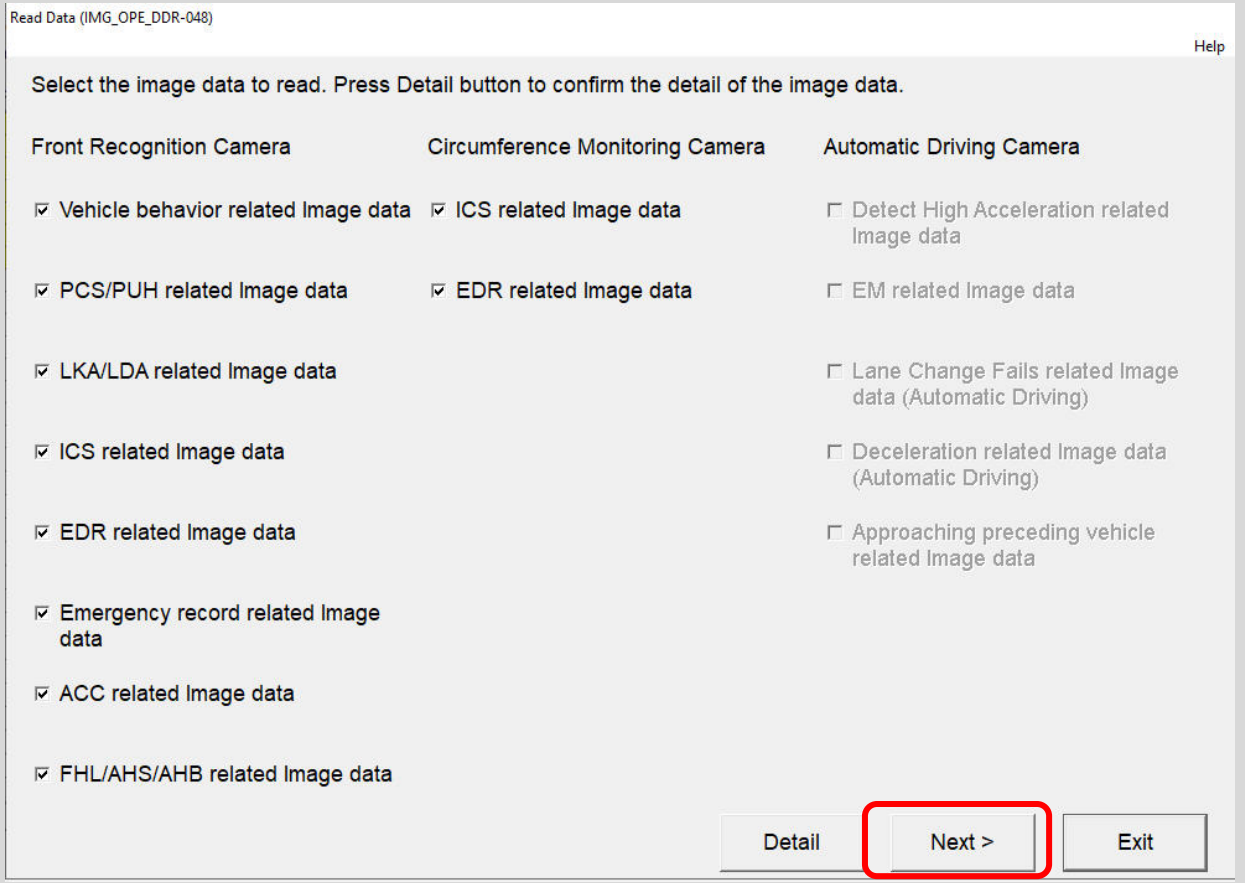
3rd Generation TSS



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•• PROTECTED 関係者外秘 PROTÉGÉ

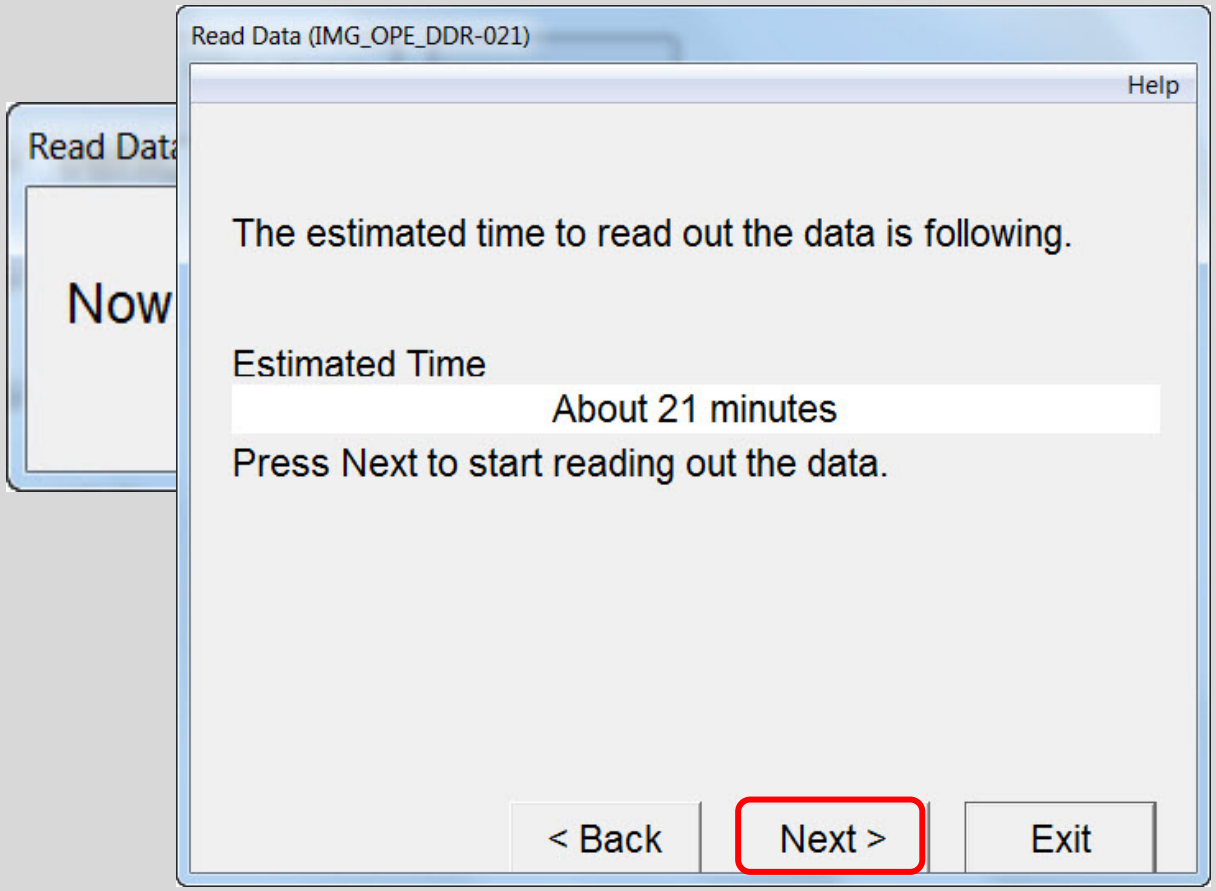
4th Generation TSS



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•• PROTECTED 関係者外秘 PROTÉGÉ

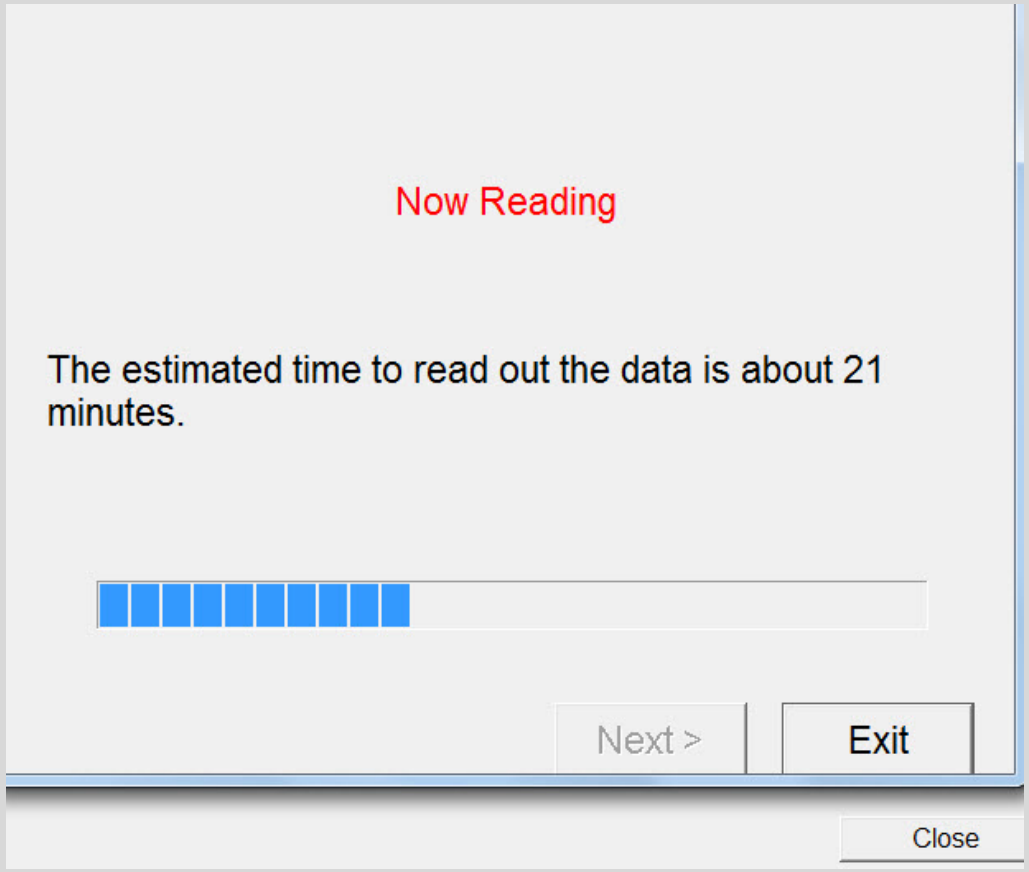
Accessing Data with Toyota Techstream



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•• PROTECTED 関係者外秘 PROTÉGÉ

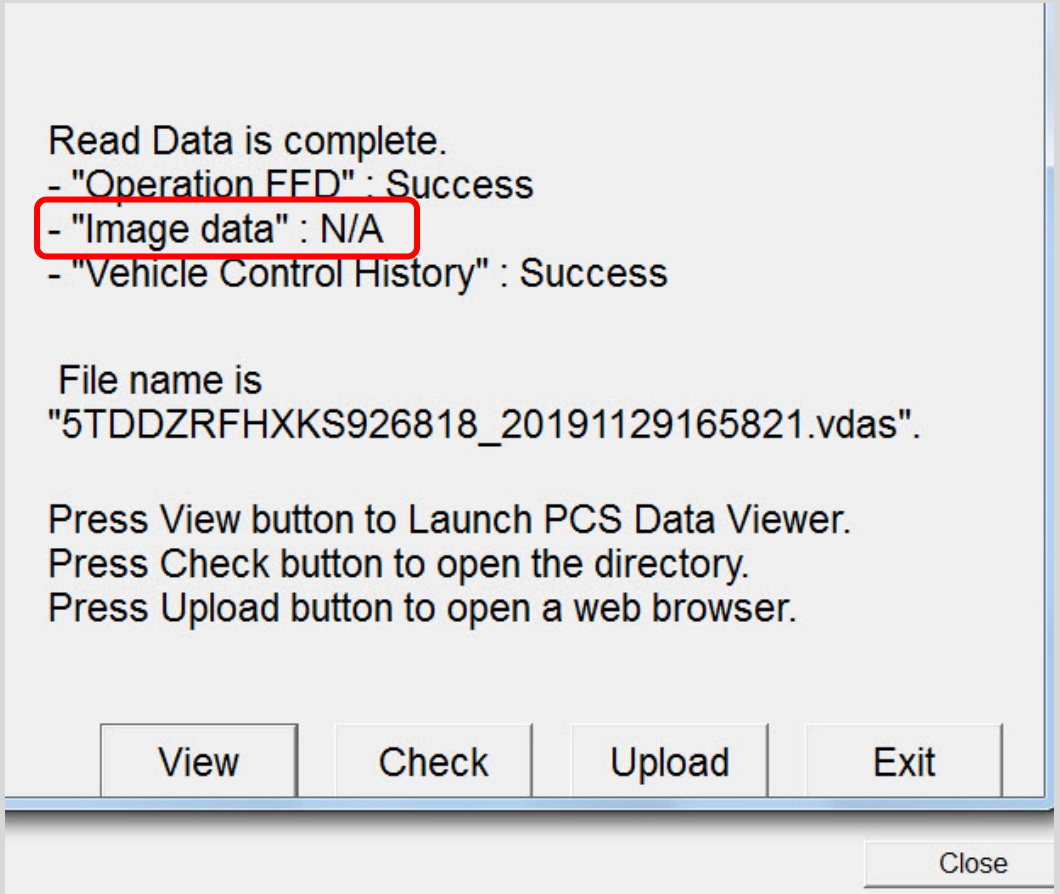
Accessing Data with Toyota Techstream



WREX 2023 – April 17-21, 2023

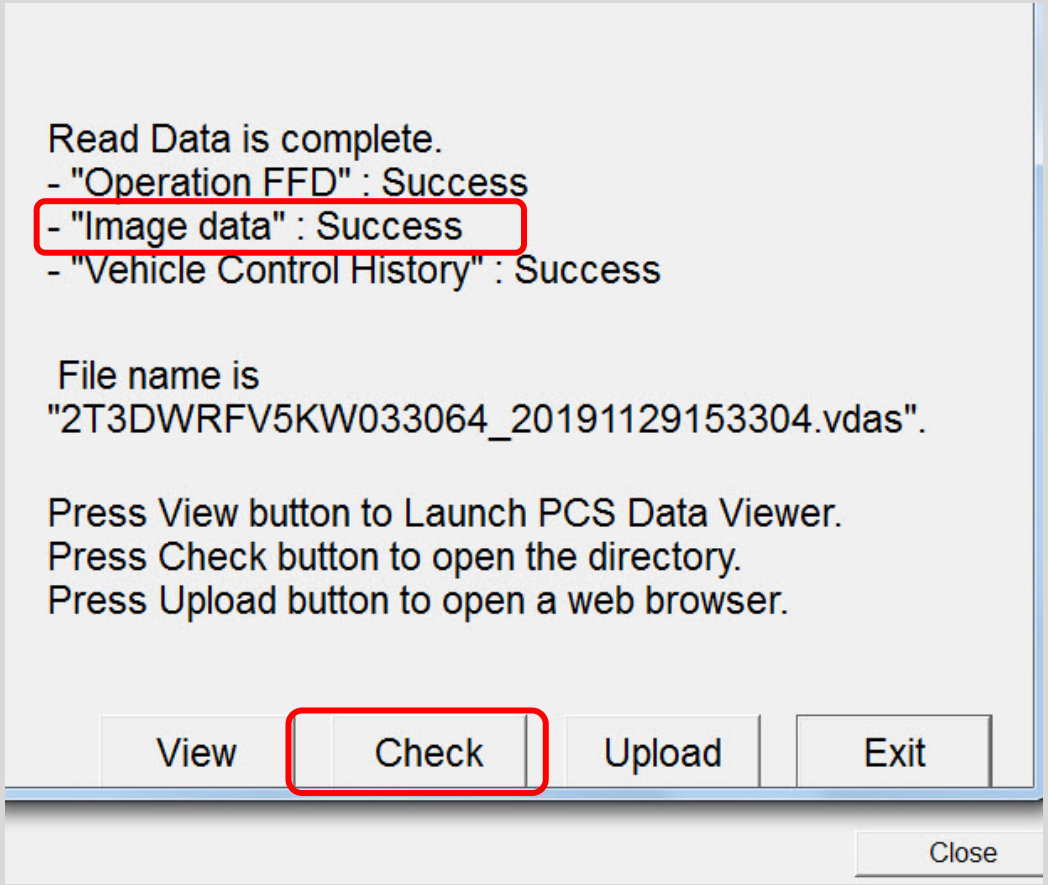
•• PROTECTED 関係者外秘 PROTÉGÉ

Accessing Data with Toyota Techstream



•• PROTECTED 関係者外秘 PROTÉGÉ

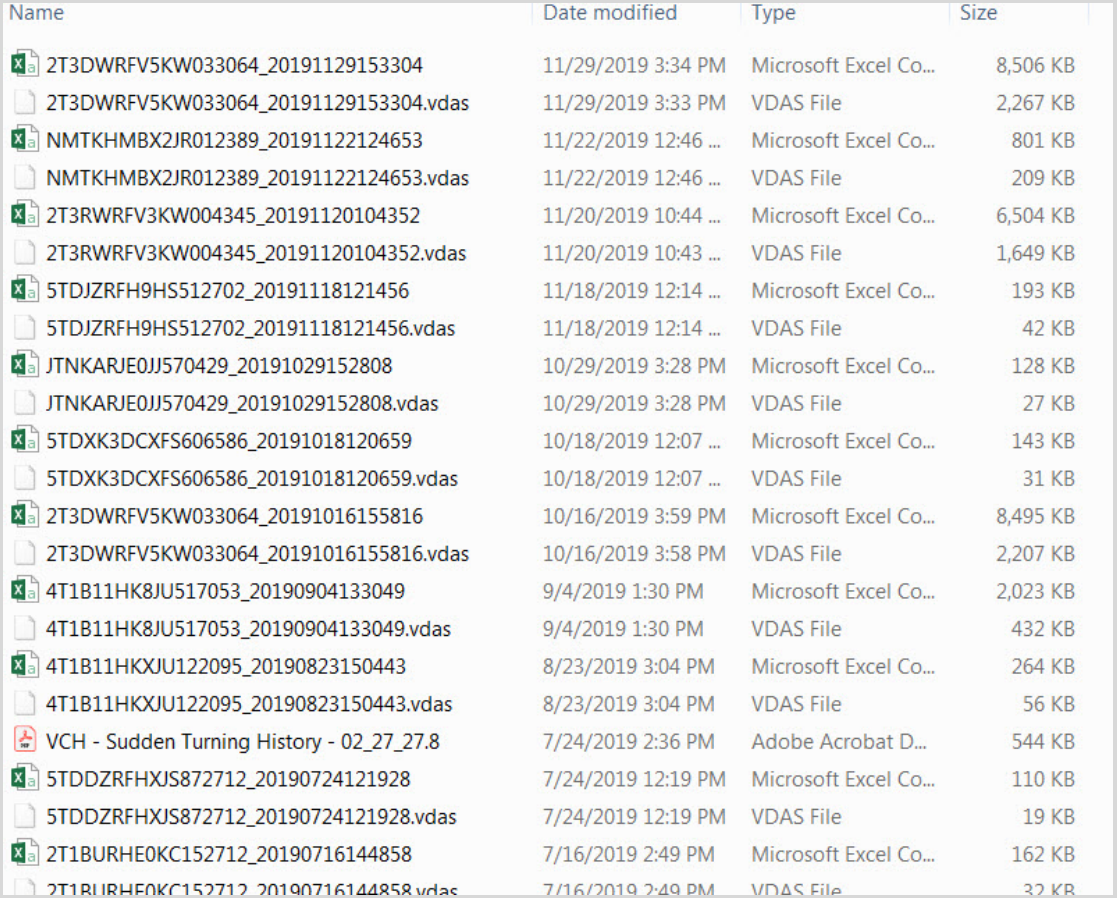
Accessing Data with Toyota Techstream



•• PROTECTED 関係者外秘 PROTÉGÉ

Accessing Data with Toyota Techstream

vdas files are saved into PCS data directory
C:\Users\Public\Documents\Techstream\PCSdata

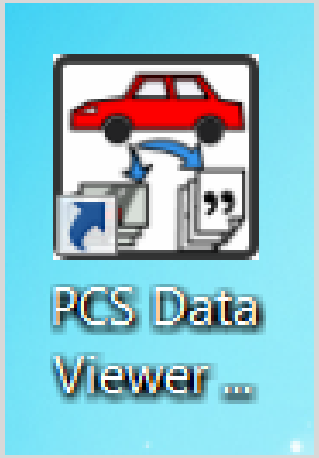


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NMTKHMBX2JR012389_20191122124653.vdas	11/22/2019 12:46 ...	VDAS File	209 KB
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VCH - Sudden Turning History - 02_27_27.8	7/24/2019 2:36 PM	Adobe Acrobat D...	544 KB
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•• PROTECTED 関係者外秘 PROTÉGÉ

PCS Data Viewer

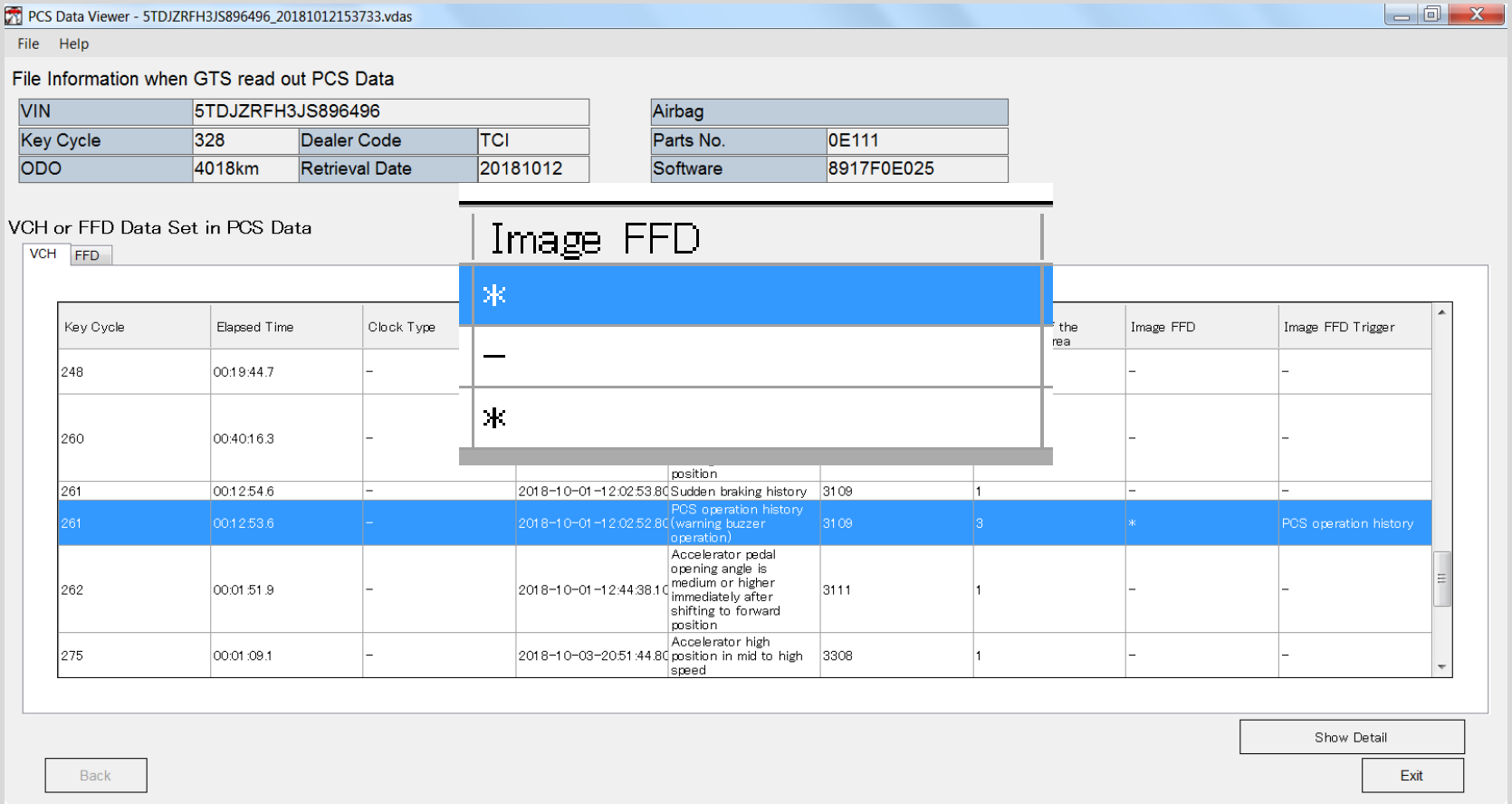
C:\Program Files (x86)\Toyota Diagnostics\PCS Data Viewer\PCS Data Viewer.exe



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•• PROTECTED 関係者外秘 PROTÉGÉ

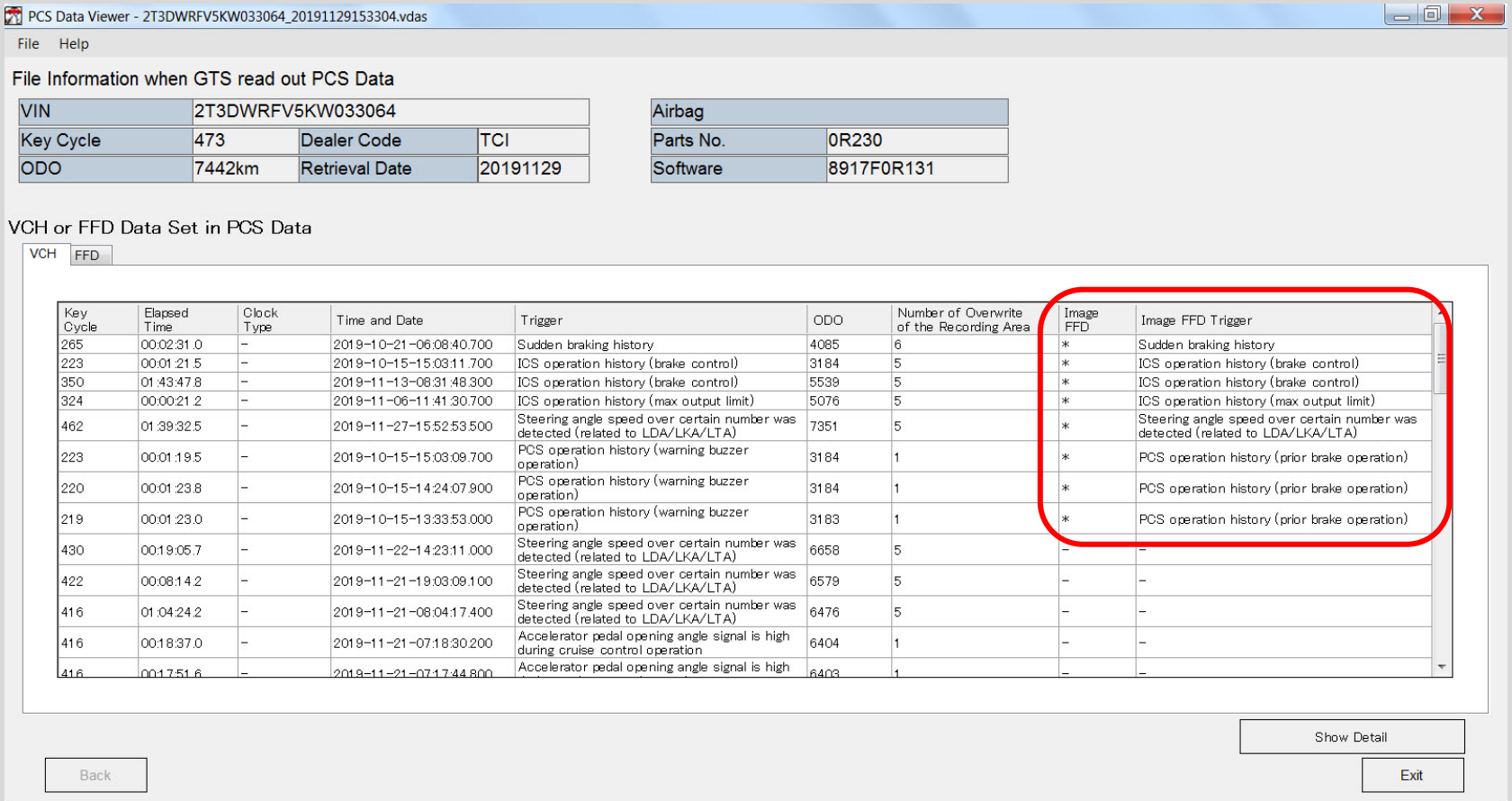
PCS Data Viewer



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•• PROTECTED 関係者外秘 PROTÉGÉ

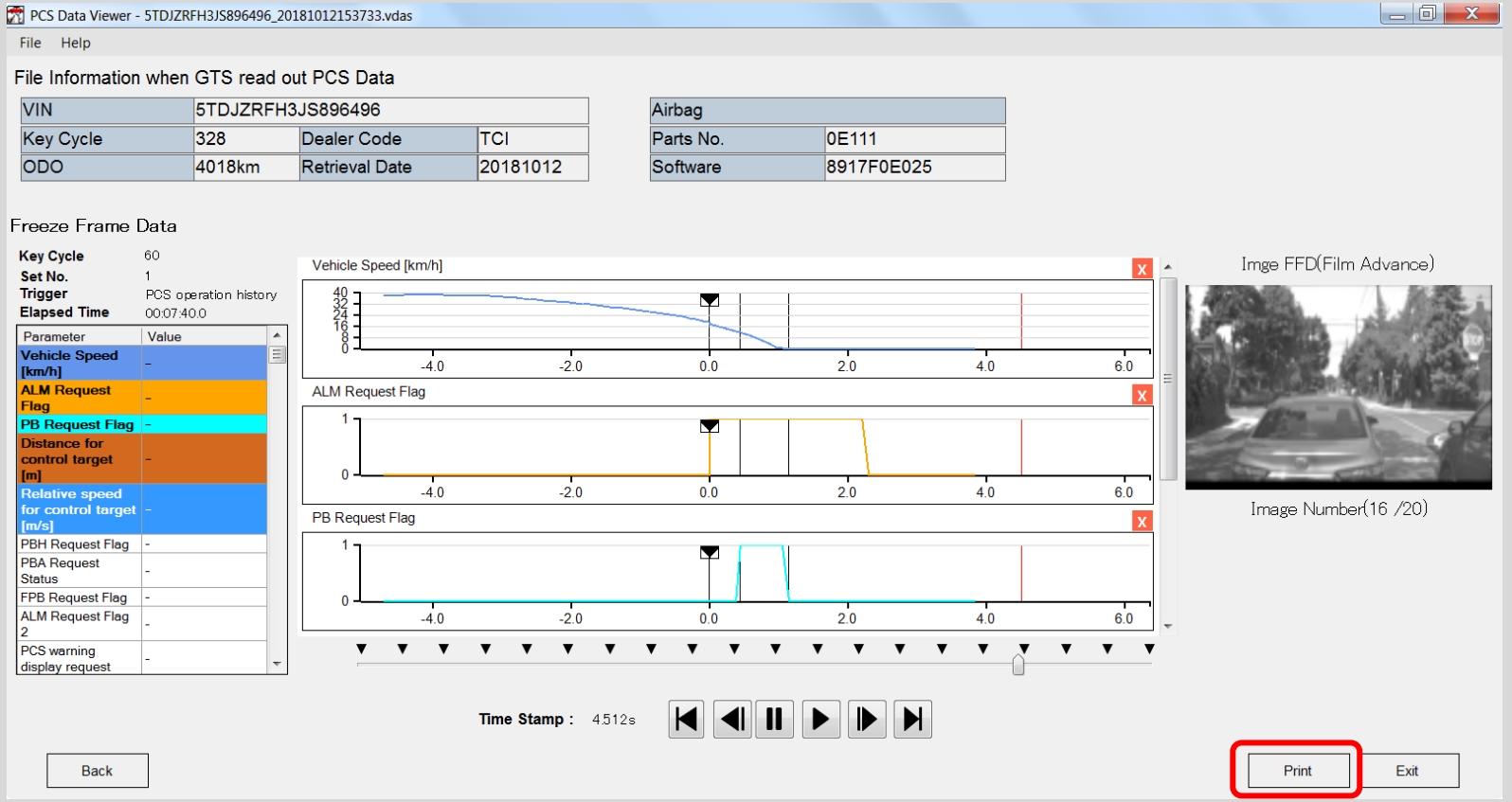
VCH / FFD with Images



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•• PROTECTED 関係者外秘 PROTÉGÉ

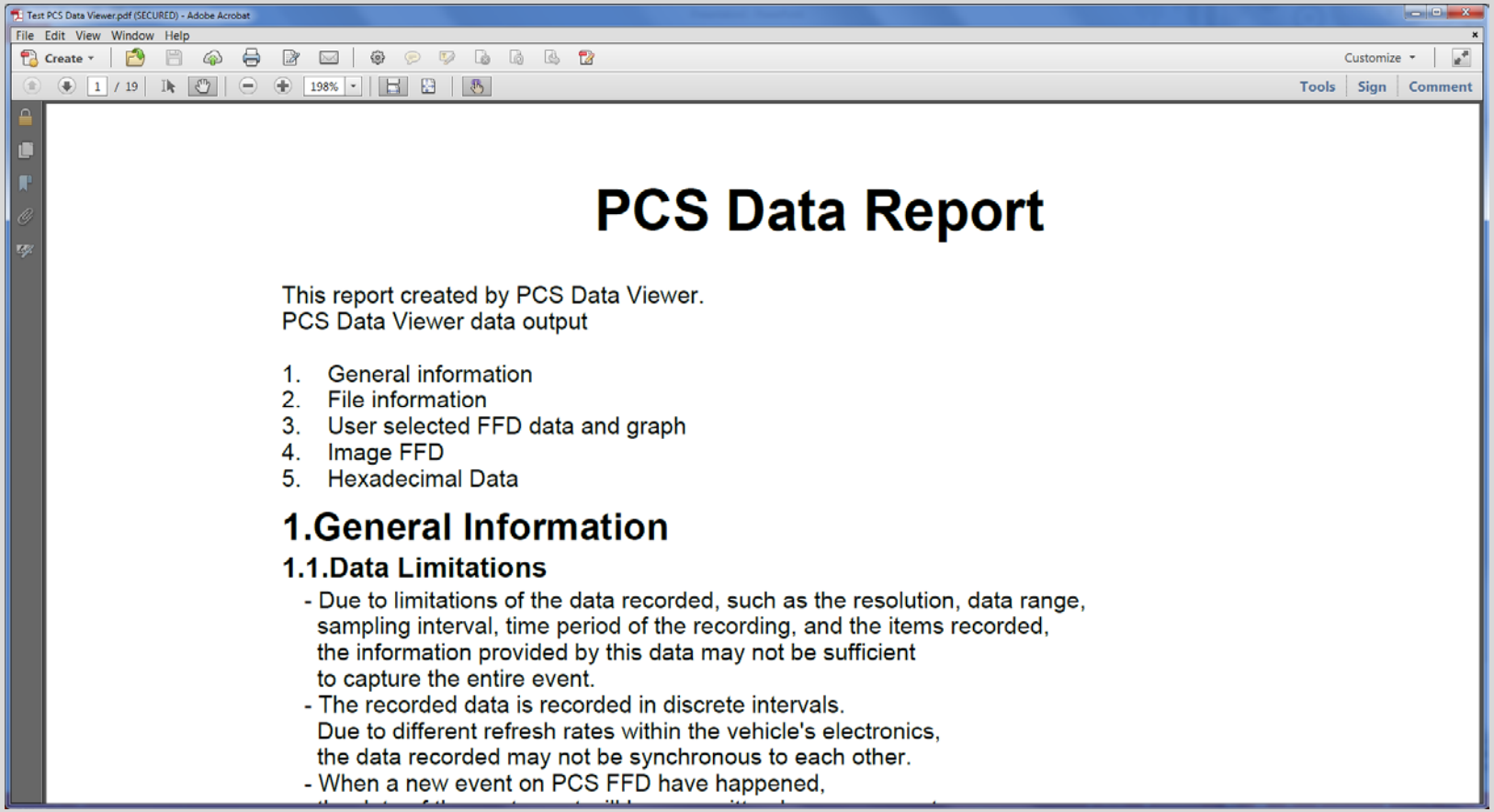
VCH / FFD with Images



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•• PROTECTED 関係者外秘 PROTÉGÉ

VCH / FFD with Images



WREX 2023 – April 17-21, 2023

•• PROTECTED 関係者外秘 PROTÉGÉ

VCH / FFD with Images

3.User Selected VCH or FFD Data

Trigger Name:Acceleration above certain amount is detected

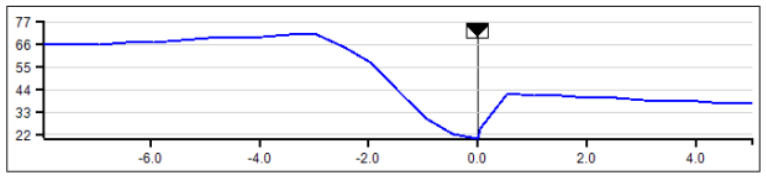
Elapsed Time: 00:09:29.2

3.1 Vehicle Speed [km/h]

Time Stamp	Parameter Value
-7.950	66
-7.450	66
-6.950	66
-6.450	67
-5.950	67
-5.450	68
-4.950	69
-4.450	69
-3.950	70
-3.450	71
-2.950	71
-2.450	65
-1.950	57
-1.450	43
-0.950	30
-0.450	22
0.000	20
0.050	25
0.550	42
1.050	41
1.550	41
2.050	40
2.550	40
3.050	39
3.550	38
4.050	38
4.550	37
5.050	37

Time Stamp	Parameter Value

Vehicle Speed [km/h]



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•• PROTECTED 関係者外秘 PROTÉGÉ

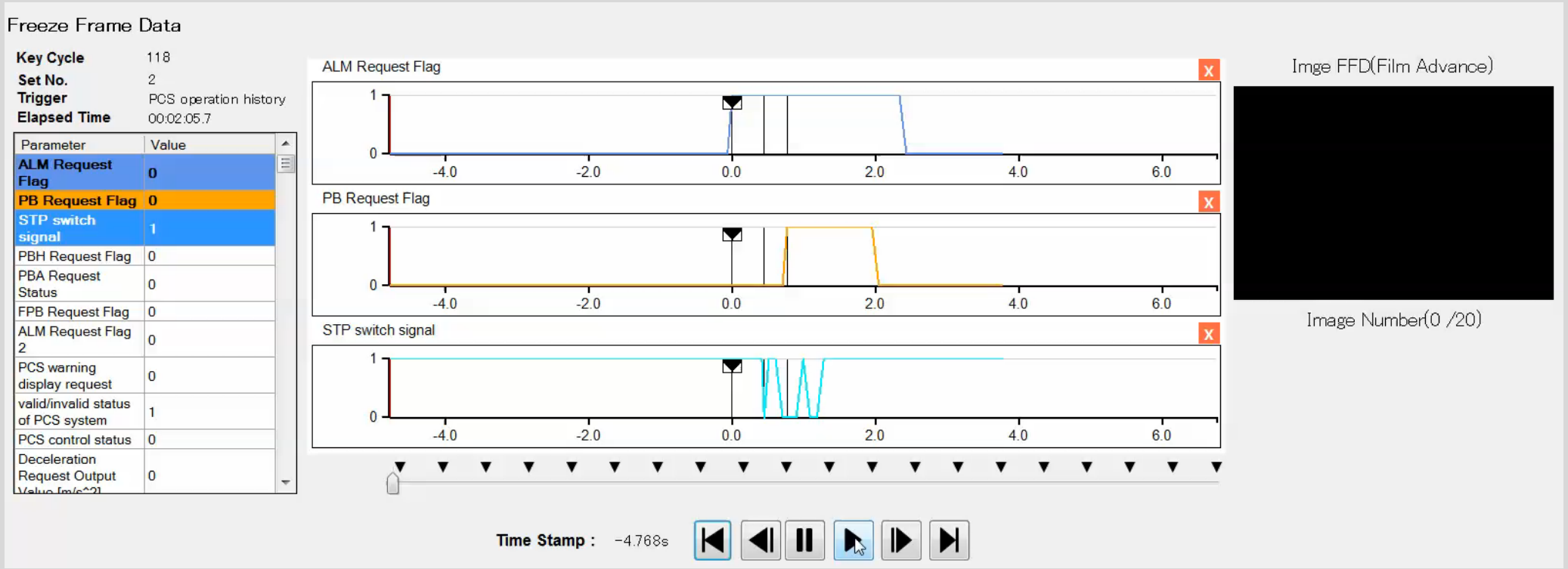
VCH / FFD with Images



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PCS Event with Pre-Braking and Impact



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Collision Related Event



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TSS with Pedestrian Detection



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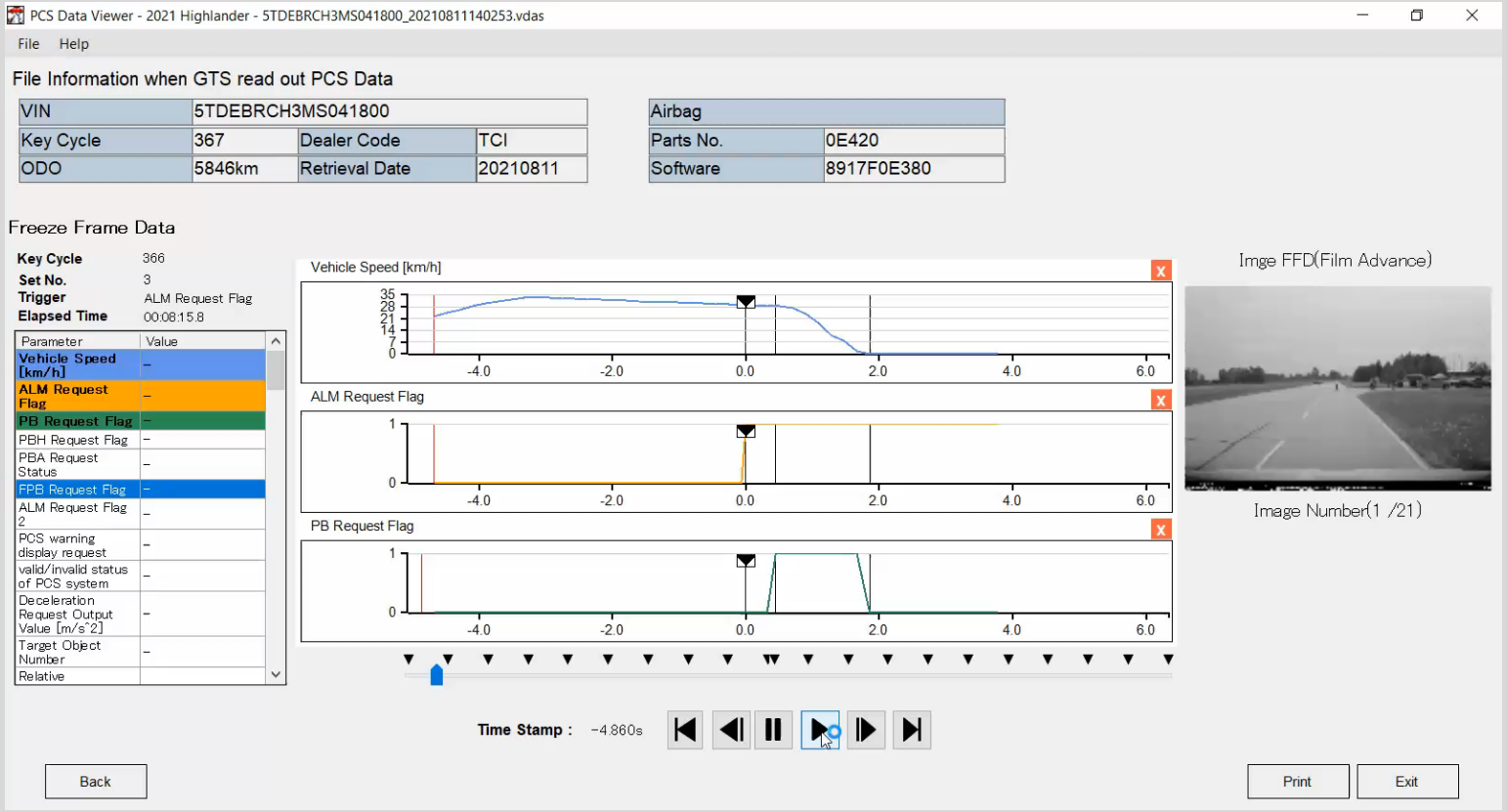
TSS with Pedestrian Detection



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TSS with Pedestrian Detection



WREX 2023 – April 17-21, 2023

•• PROTECTED 関係者外秘 PROTÉGÉ

Thank you!



Photo by Wes Vandiver

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